communication networks laboratory

Telecommunication Networks

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Based on the slides of Laurent Vanbever.
Further inspiration: Scott Shenker & Jennifer Rexford & Phillipa Gill

This week

How does communication happen?

How do we characterize it?

Briefly…

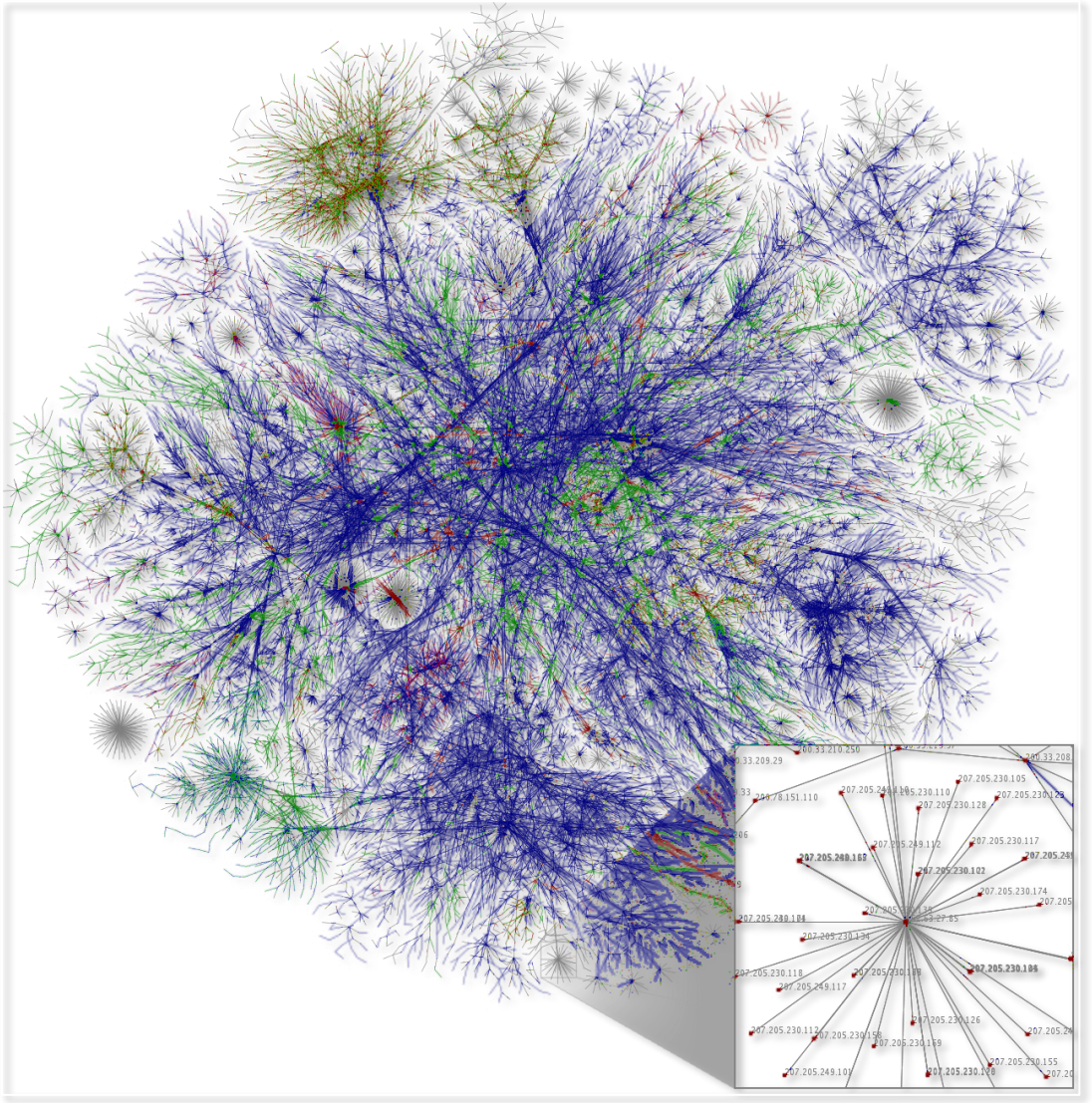
The Internet should allow

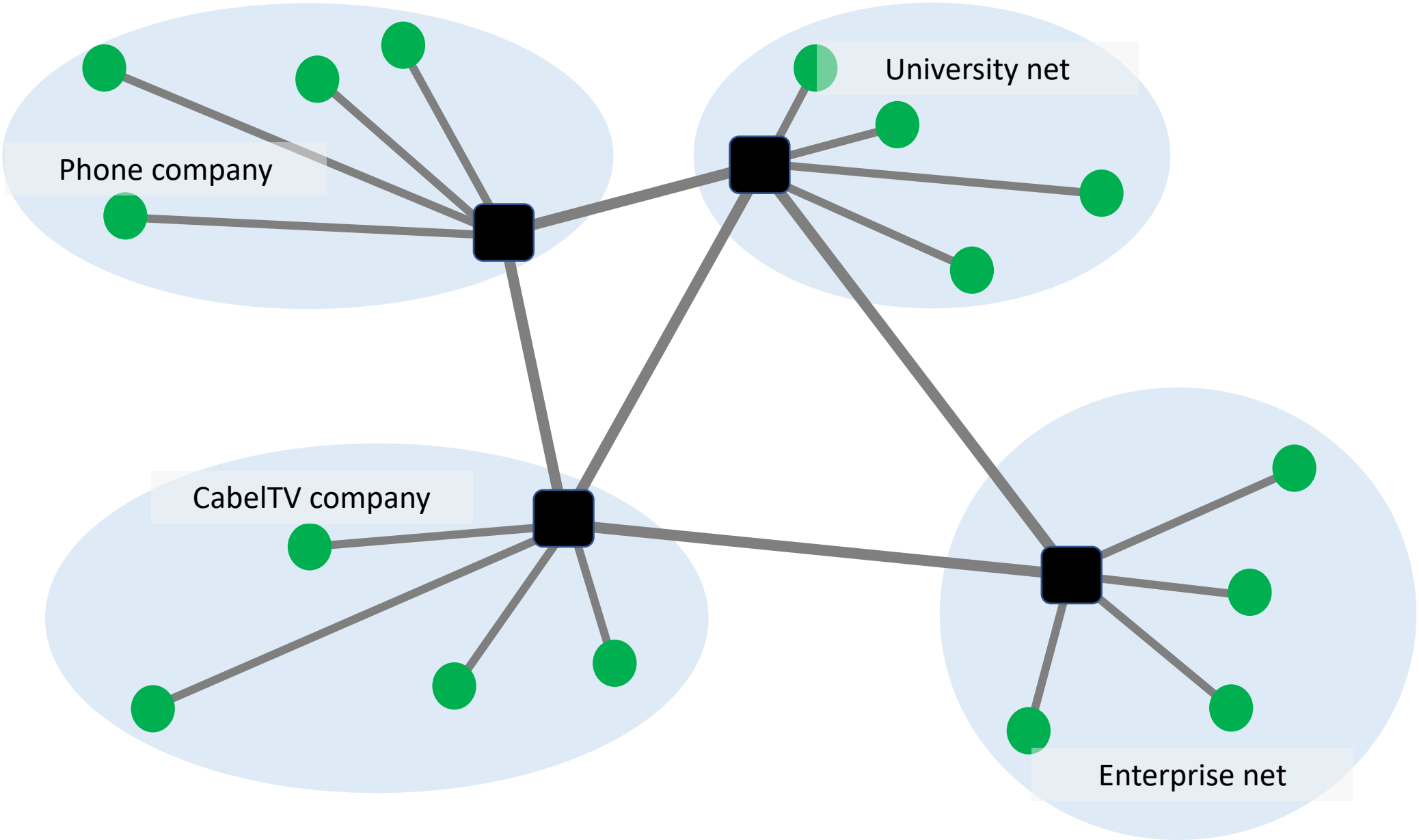
processes on different hosts

to exchange data

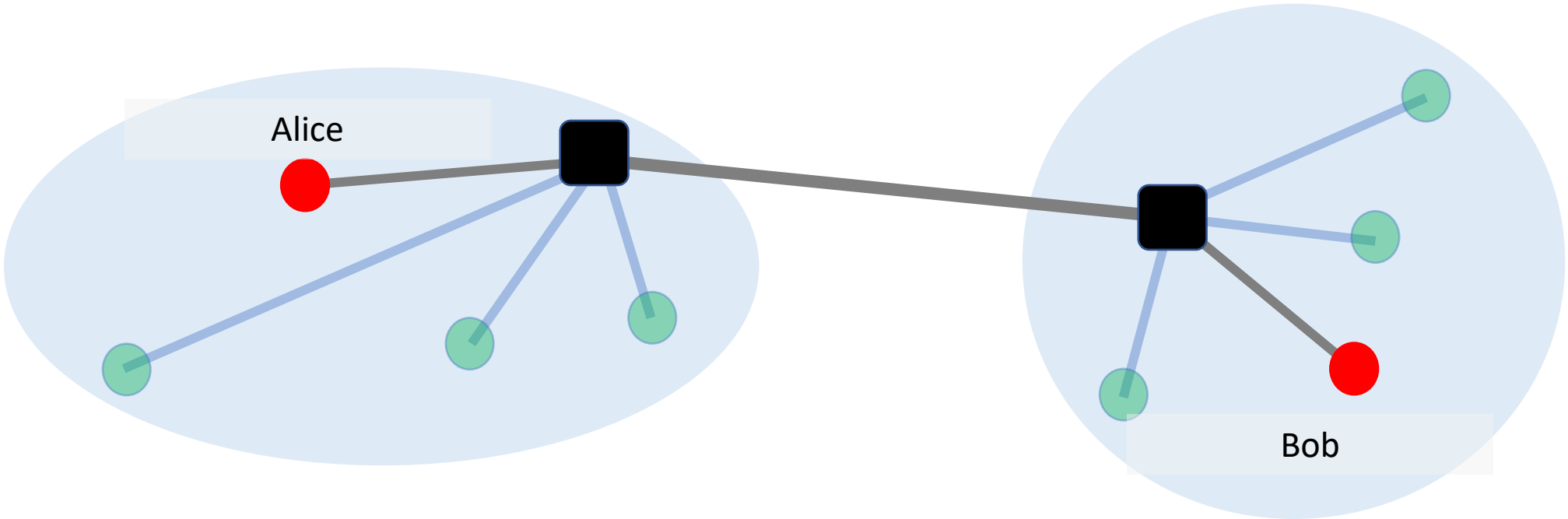
everything else is just commentary…

Ok, but how to do that in a complex system like the Internet?




Phone company
University net
CabelTV company
Enterprise net

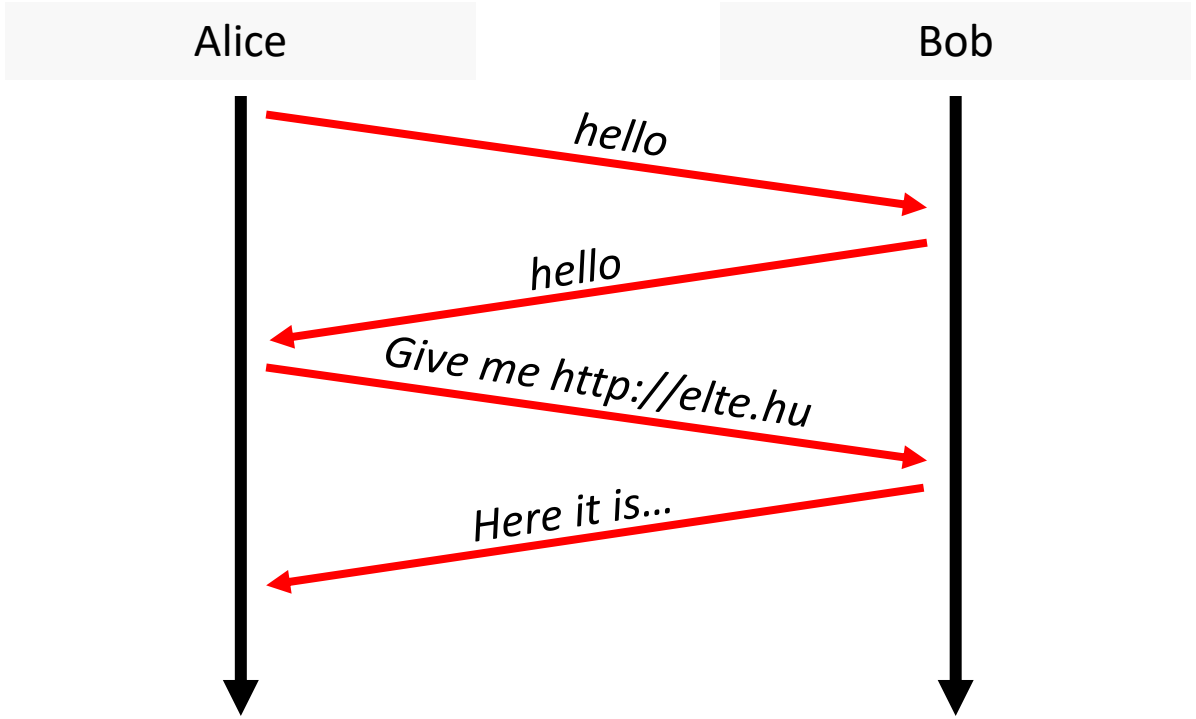
To exchange data, Alice and Bob use
a set of network protocols



A protocol is like a conversational convention

The protocol defines the order and rules the parties should follow

Who should talk next and how to respond…



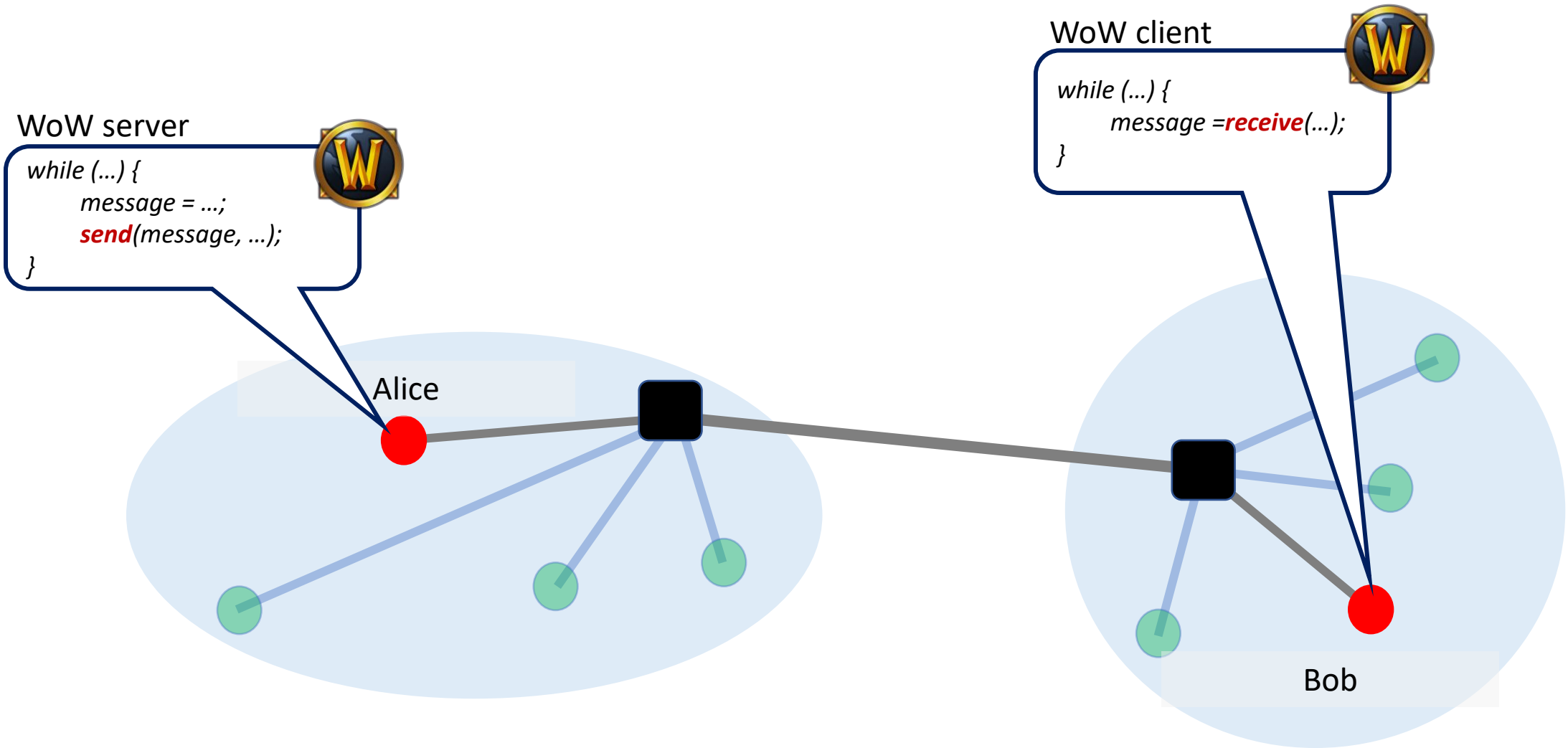
There are other kind of implementations…

Gimme, gimme, gimme
a web site after Midnight



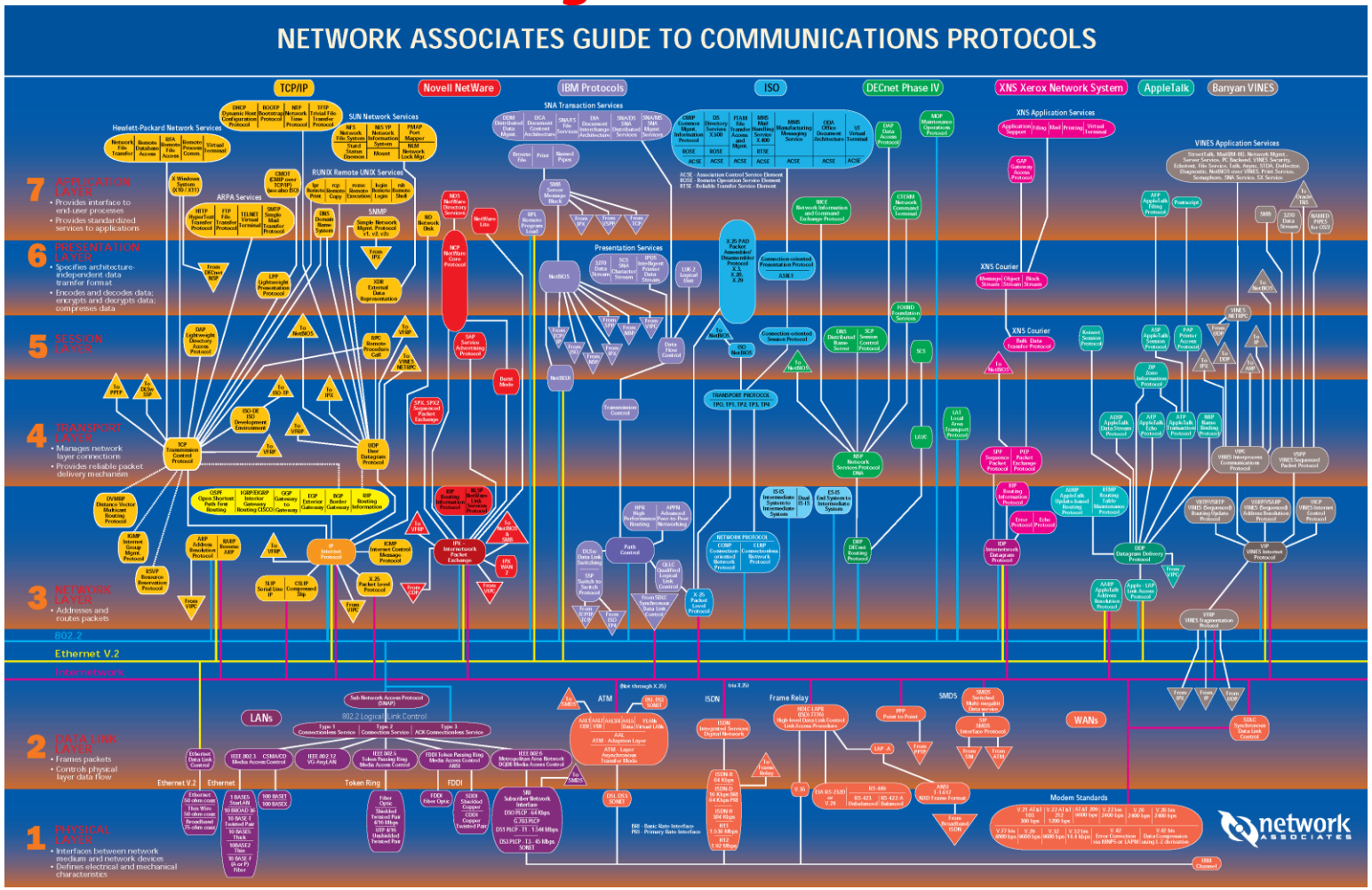


Each protocol is governed by a specific API



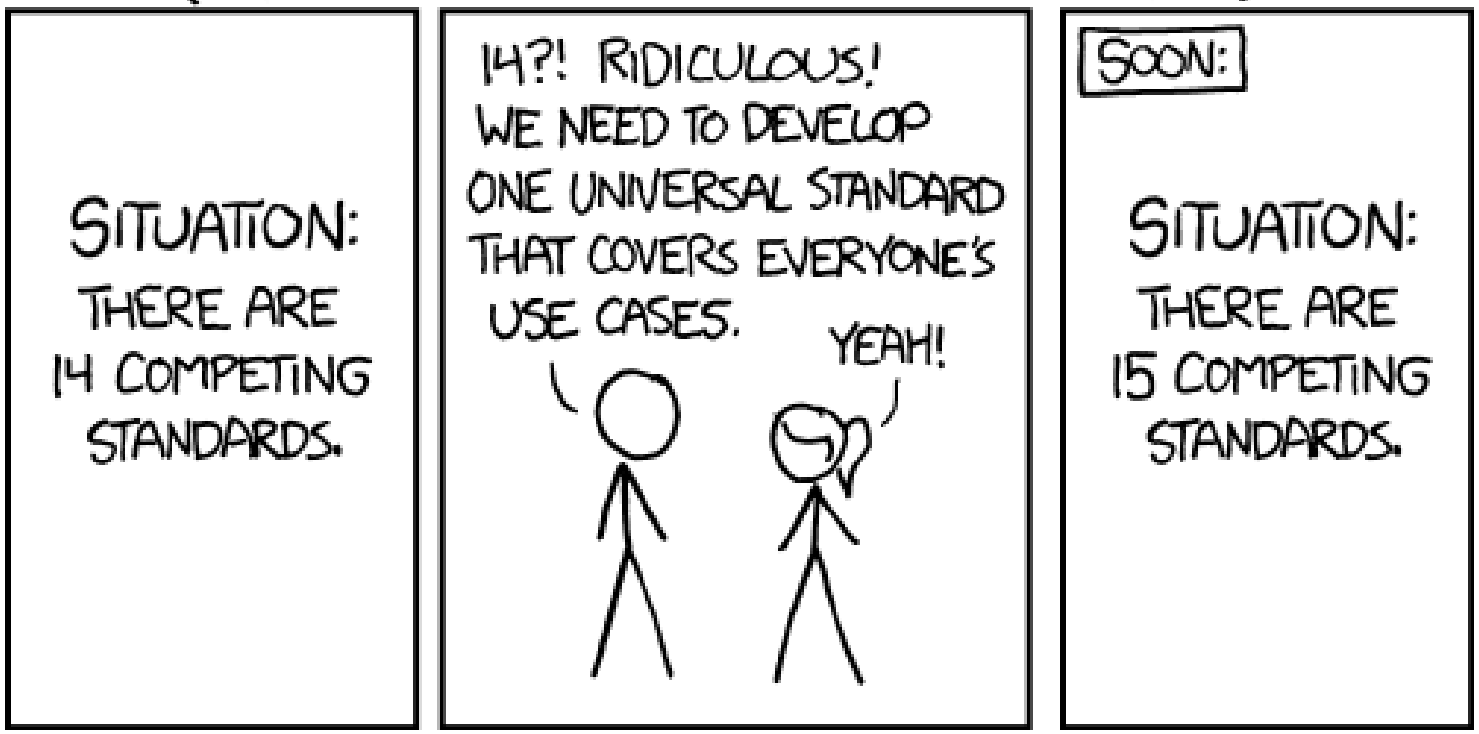
In practice, many existing protocols…

*How does the Internet **organize this**???*



HOW STANDARDS PROLIFERATE:

(SEE: A/C CHARGERS, CHARACTER ENCODINGS, INSTANT MESSAGING, ETC.)



Modularity is a key component of any good system

Problem

can't build large systems out of spaghetti code

hard (if not, impossible) to understand, debug, update

need to bound the scope of changes

evolve the system without rewriting it from scratch

Solution

Modularity is how we do it

…and understand the system at a higher-level



„Modularity,
based on abstraction,
is **the way** things get done”

Barbara Liskov, MIT

To provide structure to the design of network protocols, network designers organize protocols in layers

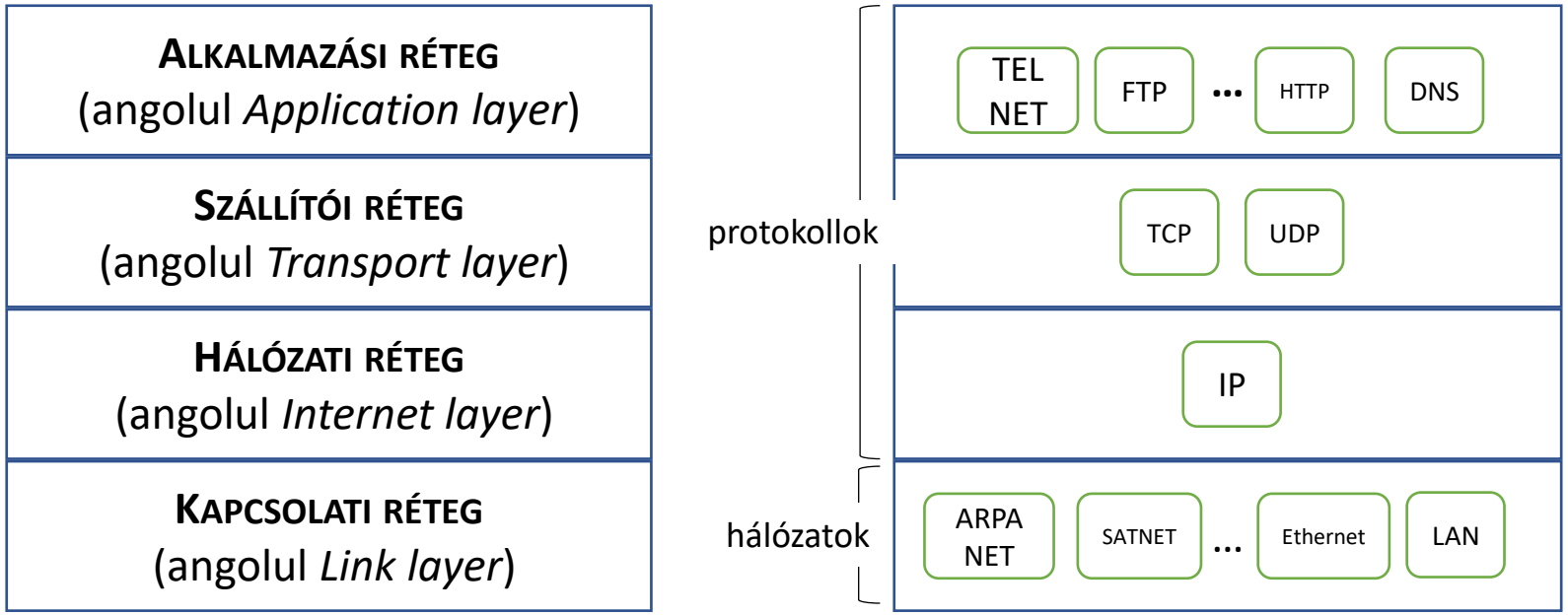
and the network hardware/software that implement them

Hálózatok modelljei

- Internet rétegmodelljei
  - TCP/IP modell: 4 réteget különböztet meg. 1982 márciusában az amerikai hadászati célú számítógépes hálózatok standardja lett. 1985-től népszerűsítették kereskedelmi felhasználásra. (*Interop*)
  - Hibrid TCP/IP modell: 5 réteget különböztet meg (*Tanenbaum, Stallings, Kurose, Forouzan*)

- Nyílt rendszerek hálózatának standard modellje
  - ***O**pen **S**ystem **I**nterconnection Reference Model*: Röviden OSI referencia modell, amely egy 7-rétegű standard, koncepcionális modellt definiál kommunikációs hálózatok belső funkcionalitásaihoz. (*ISO/IEC 7498-1*)

TCP/IP modell (RFC 1122)



TCP/IP modell rétegei („bottom-up”)

- **Kapcsolati réteg / Host-to-network or Link layer**
  - nem specifikált
  - a LAN-tól függ
- **Internet réteg / Internet or Network layer**
  - speciális csomagformátum
  - útvonal meghatározás (routing)
  - csomag továbbítás (angolul *packet forwarding*)
- **Szállítói réteg / Transport layer**
  - **T**ransport **C**ontrol **P**rotocol
    - megbízható, kétirányú bájt-folyam átviteli szolgáltatás
    - szegmentálás, folyamfelügyelet, multiplexálás
  - **U**ser **D**atagram **P**rotocol
    - nem megbízható átviteli szolgáltatás
    - nincs folyamfelügyelet
- **Alkalmazási réteg / Application layer**
  - Szolgáltatások nyújtása: Telnet, FTP, SMTP, HTTP, NNTP, DNS, SSH, etc.

Rétegek jellemzése



- Szolgáltatás
  - Mit csinál az adott réteg?
- Interfész
  - Hogyan férhetünk hozzá a réteghez?
- Protokoll
  - Hogyan implementáljuk a réteget?

Fizikai réteg



- Szolgáltatás
  - Információt visz át két fizikailag összekötött eszköz között
  - definiálja az eszköz és a fizikai átviteli közeg kapcsolatát
- Interfész
  - Specifikálja egy bit átvitelét
- Protokoll
  - Egy bit kódolásának sémája
  - Feszültség szintek
  - Jelek időzítése
- Példák: koaxiális kábel, optikai kábel, rádió frekvenciás adó

Adatkapcsolati réteg



- Szolgáltatás
  - Adatok keretekre tördelésezés: határok a csomagok között
  - Közeghozzáférés vezérlés (MAC)
  - Per-hop megbízhatóság és folyamvezérlés
- Interfész
  - Keret küldése két közös médiumra kötött eszköz között
- Protokoll
  - Fizikai címzés (pl. MAC address, IB address)
- Példák: Ethernet, Wifi, InfiniBand

Hálózati réteg



- Szolgáltatás
  - Csomagtovábbítás
  - Útvonalválasztás
  - Csomag fragmentálás kezelése
  - Csomag ütemezés
  - Puffer kezelés
- Interfész
  - Csomag küldése egy adott végpontnak
- Protokoll
  - Globálisan egyedi címeket definiálása
  - Routing táblák karbantartása
- Példák: Internet Protocol (IPv4), IPv6

Szállítói réteg



- Szolgáltatás
  - Multiplexálás/demultiplexálás
  - Torlódásvezérlés
  - Megbízható, sorrendhelyes továbbítás
- Interfész
  - Üzenet küldése egy célállomásnak
- Protokoll
  - Port szám
  - Megbízhatóság/Hiba javítás
  - Folyamfelügyelet
- Példa: UDP, TCP

Ülés v. Munkamenet réteg



- Szolgáltatás
  - kapcsolat menedzsment: felépítés, fenntarás és bontás
  - munkamenet típusának meghatározása
  - szinkronizációs pont menedzsment (checkpoint)
- Interfész
  - Attól függ…
- Protokoll
  - Token menedzsment
  - Szinkronizációs checkpoints beszúrás
- Példa: nincs

Megjelenítési réteg



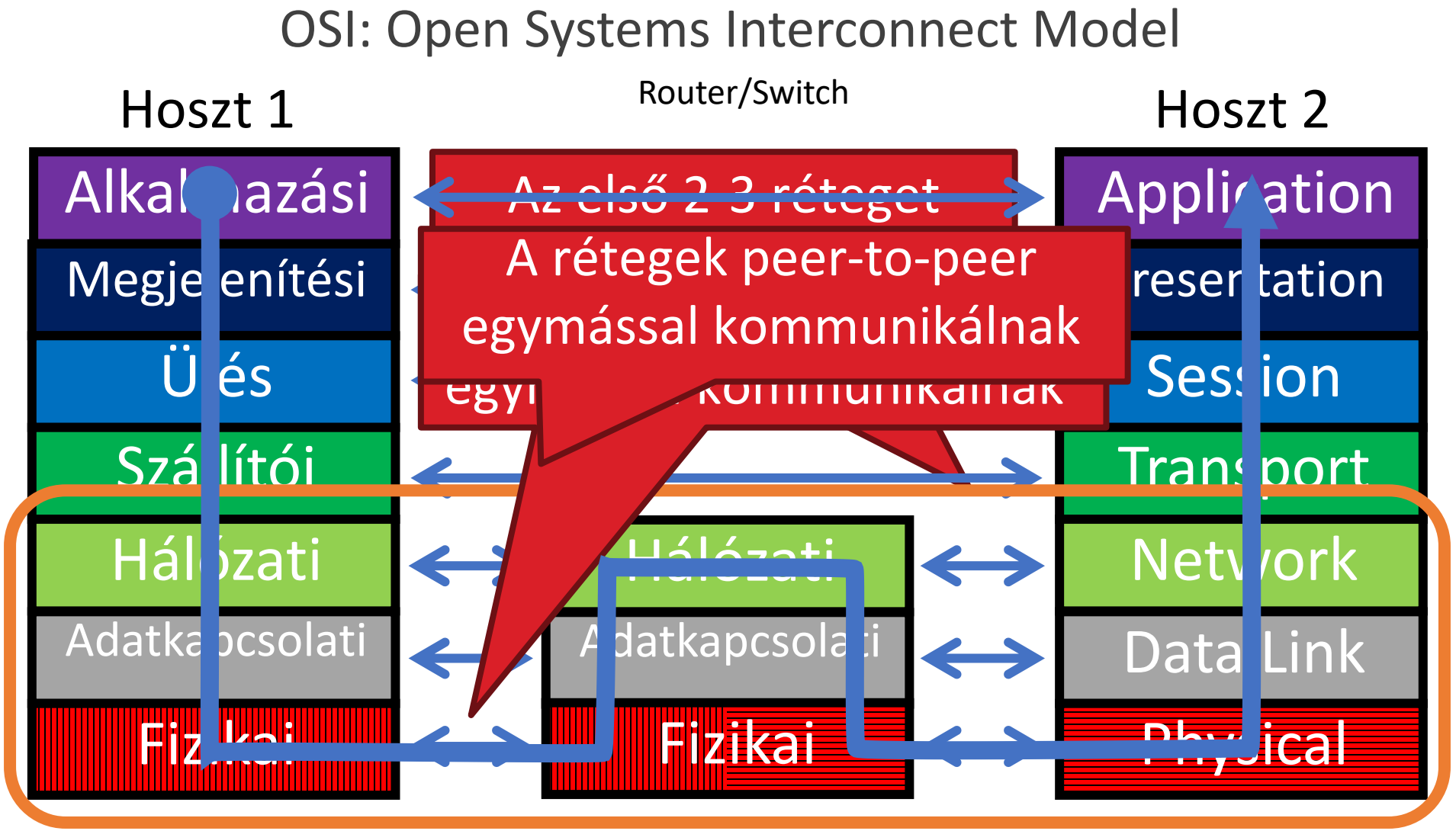
- Szolgáltatás
  - Adatkonverzió különböző reprezentációk között
  - Pl. big endian to little endian
  - Pl. Ascii to Unicode
- Interfész
  - Attól függ…
- Protokoll
  - Adatformátumokat definiál
  - Transzformációs szabályokat alkalmaz
- Példa: nincs

Alkalmazási réteg



- Szolgáltatás
  - Bármi…
- Interfész
  - Bármi…
- Protokoll
  - Bármi…
- Példa: kapcsold be a mobilod és nézd meg milyen appok vannak rajta…

ISO OSI modell



Hybrid model – 5 layers

Each layer provides a service to the layer above

	layer	service provided
L5	**Application**	*high level network access*
L4	**Transport**	*end-to-end delivery (reliable or not)*
L3	**Network**	*global best-effort delivery*
L2	**Link**	*local best-effort delivery*
L1	**Physical**	*physical transfer of bits*

Each layer provides a service to the layer above by using the services of the layer directly below it

Applications

…built on…

Reliable (or unreliable) transport

…built on…

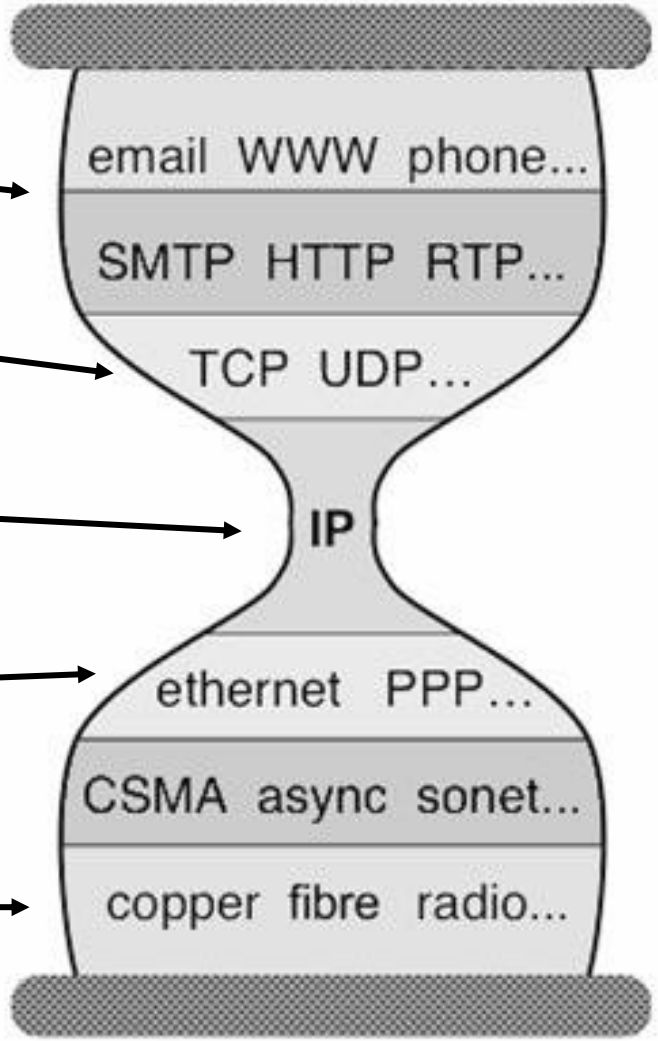
Best-effort global packet delivery

…built on…

Best-effort local packet delivery

…built on…

Physical transfer of bits



Each layer has a unit of data (aka protocol data unit)

	layer	role (PDU)
L5	**Application**	exchanges messages between processes
L4	**Transport**	transports segments between end-systems
L3	**Network**	moves packets around the network
L2	**Link**	moves frames across a link
L1	**Physical**	moves bits across a physical medium

Each layer (except for L3) is implemented with different protocols

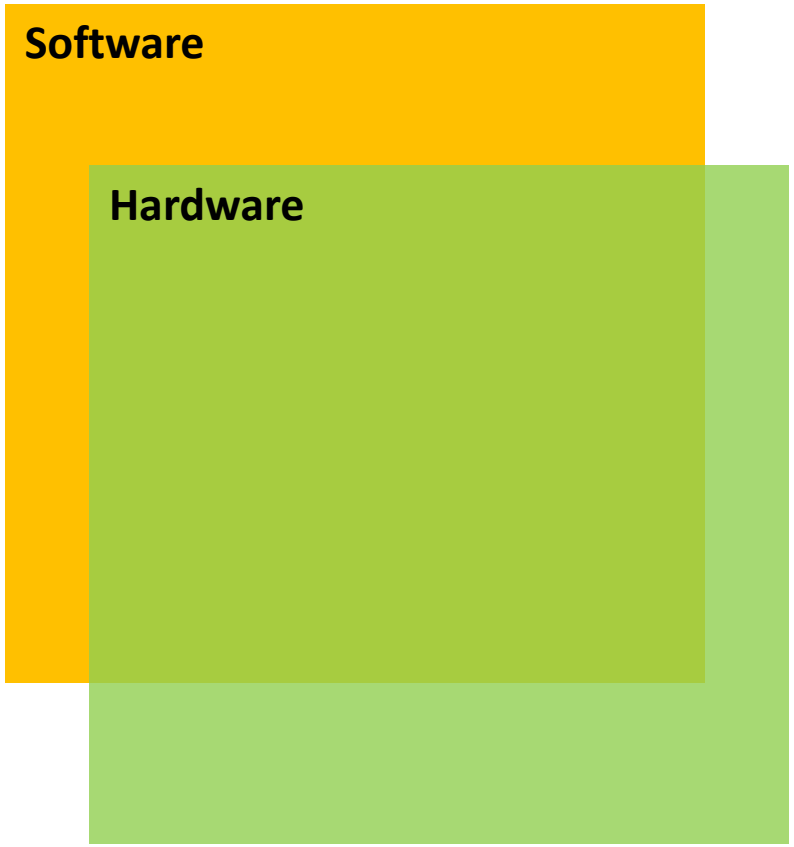
	layer	protocols
L5	**Application**	HTTP, SMTP, FTP, SIP, …
L4	**Transport**	TCP, UDP, SCTP
L3	**Network**	IP
L2	**Link**	Ethernet, Wifi, ADSL, WiMAX, LTE, …
L1	**Physical**	Twisted pair, fiber, coaxial cable, …

The Internet Protocol (IP) is the **glue** acting as a unifying network layer

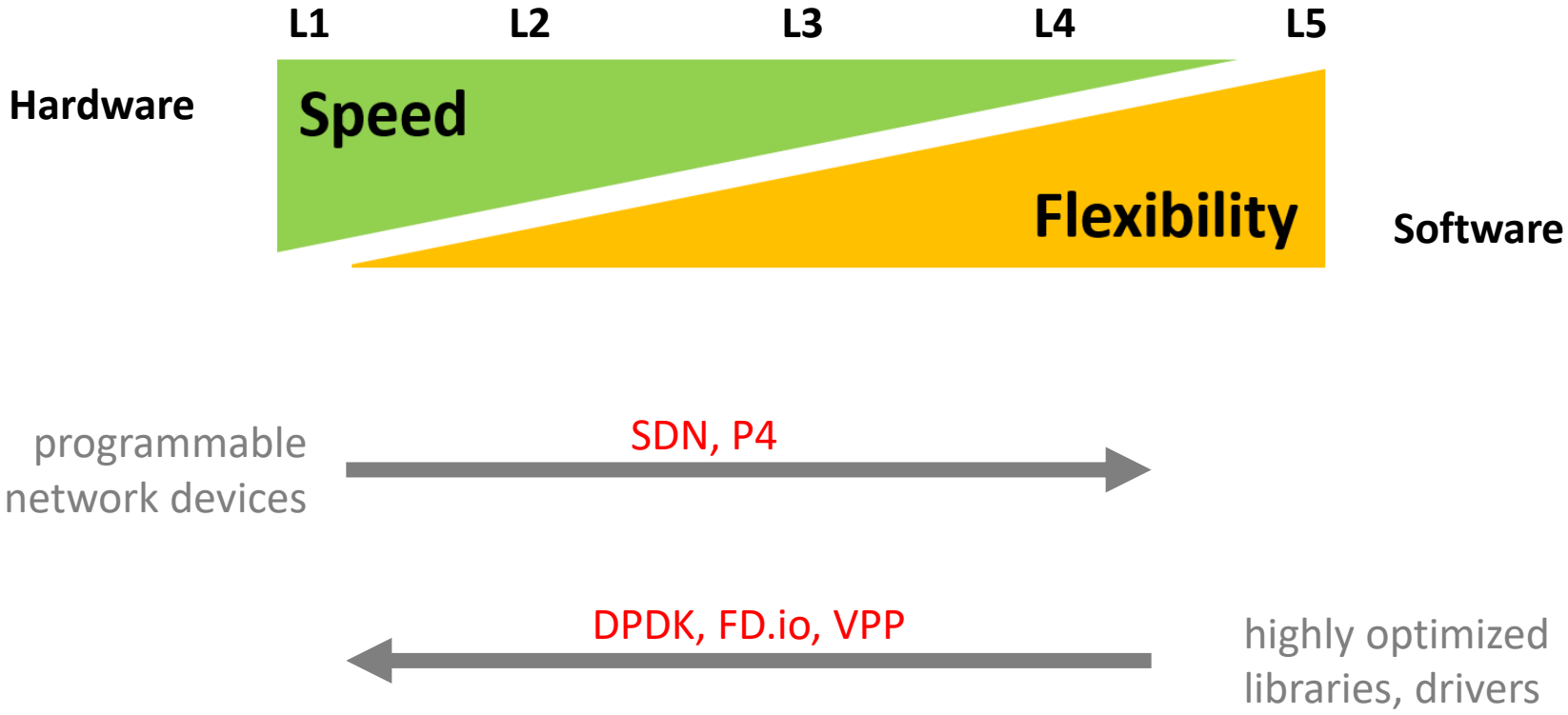
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L3	**Network**	IP
L2	**Link**	Ethernet, Wifi, ADSL, WiMAX, LTE, …
L1	**Physical**	Twisted pair, fiber, coaxial cable, …

Each layer is implemented with different protocols and technologies

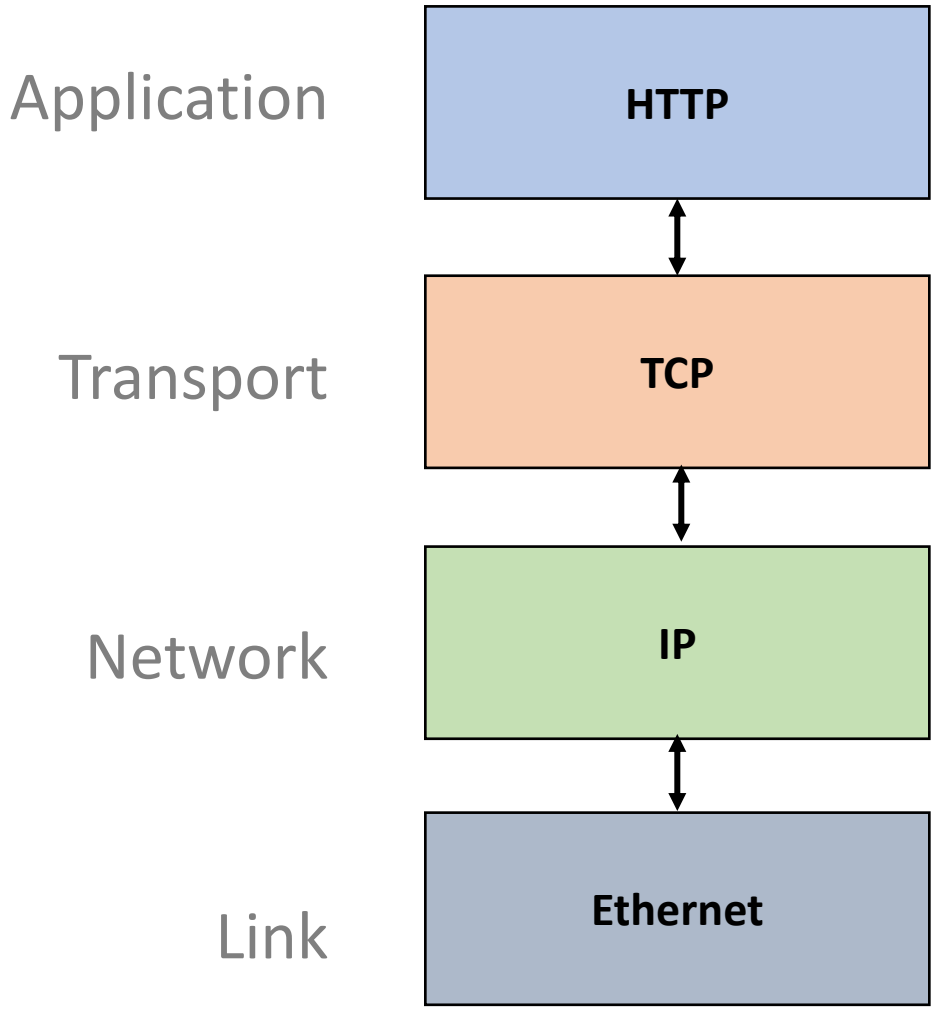
layer		technology
L5	**Application**	
L4	**Transport**	
L3	**Network**	
L2	**Link**	
L1	**Physical**	

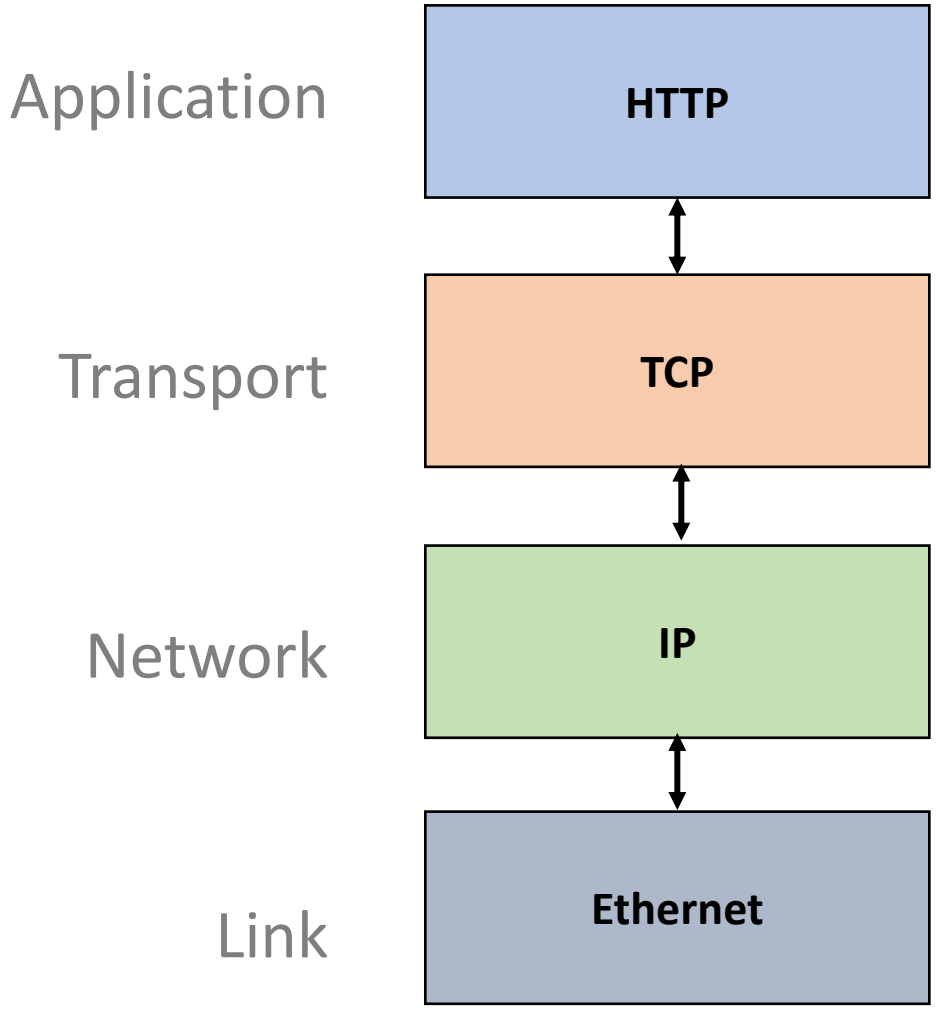


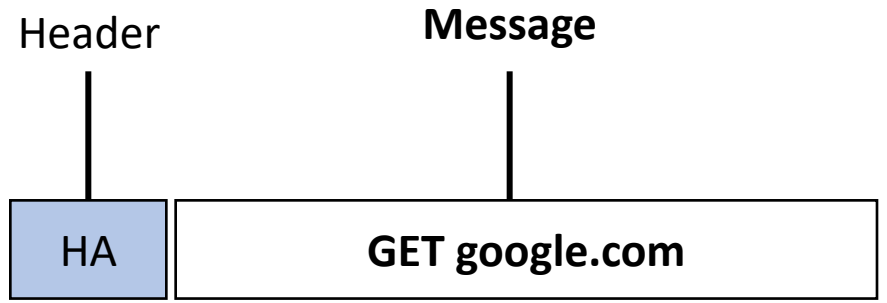
Software and hardware advancements

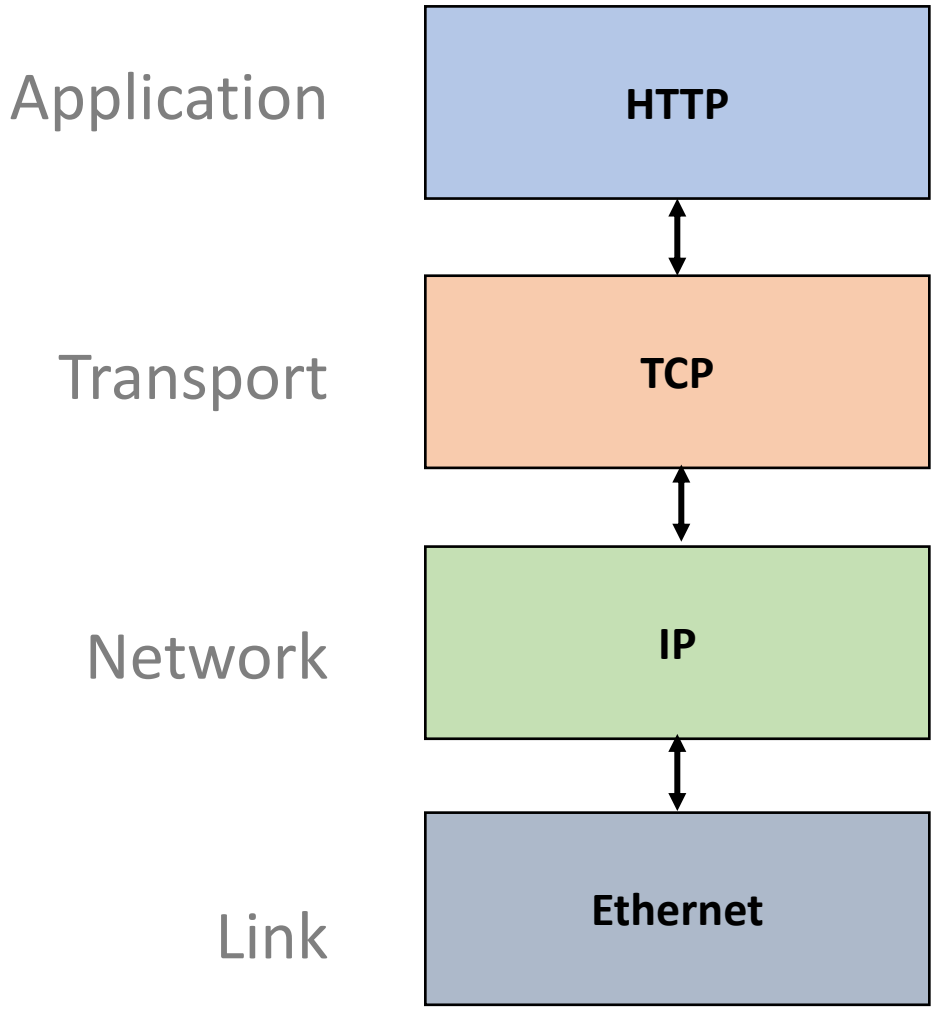


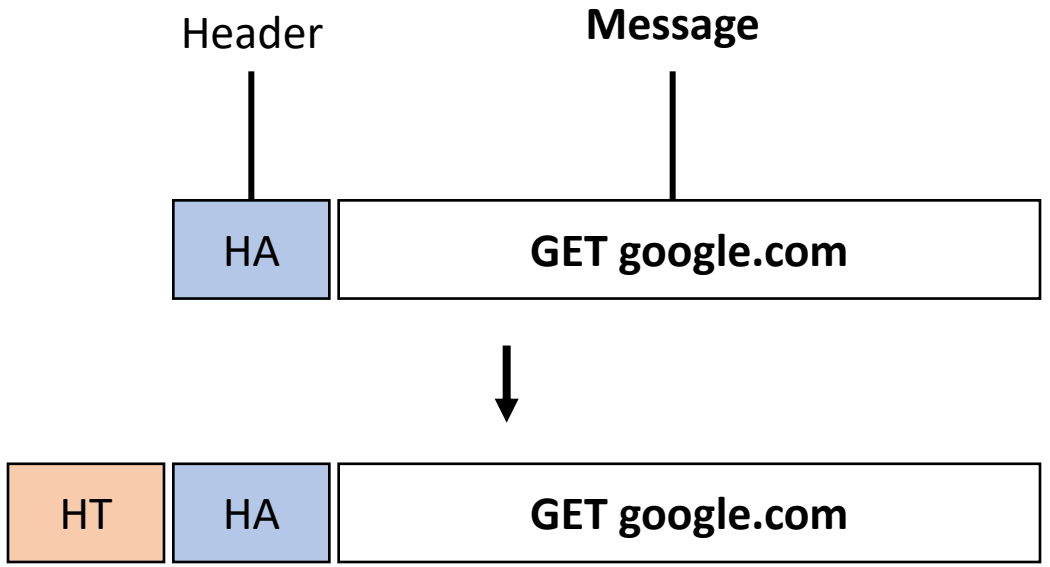
Each layer takes messages from the layer above,
and *encapsulates* with its own header and/or trailer

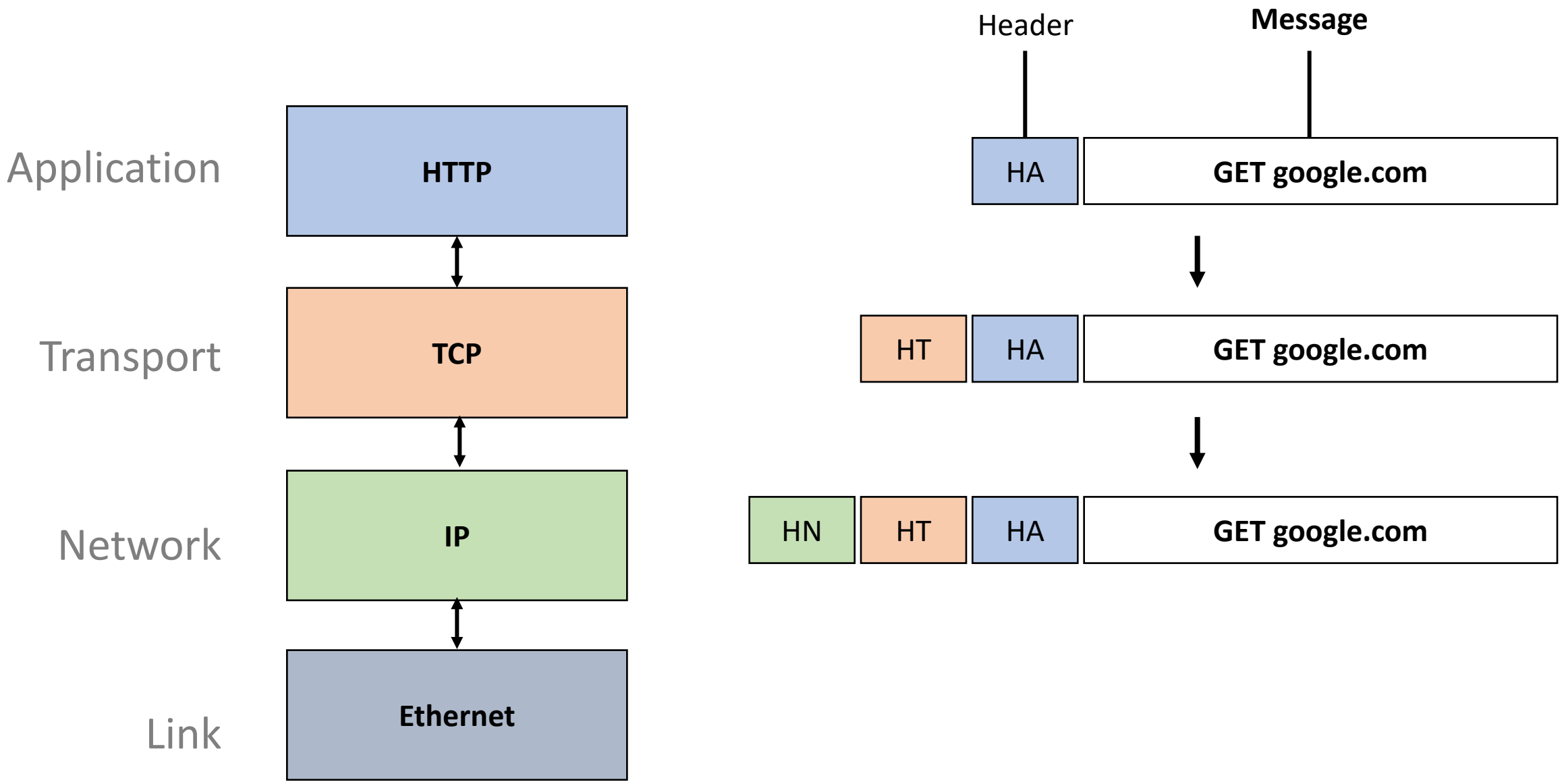


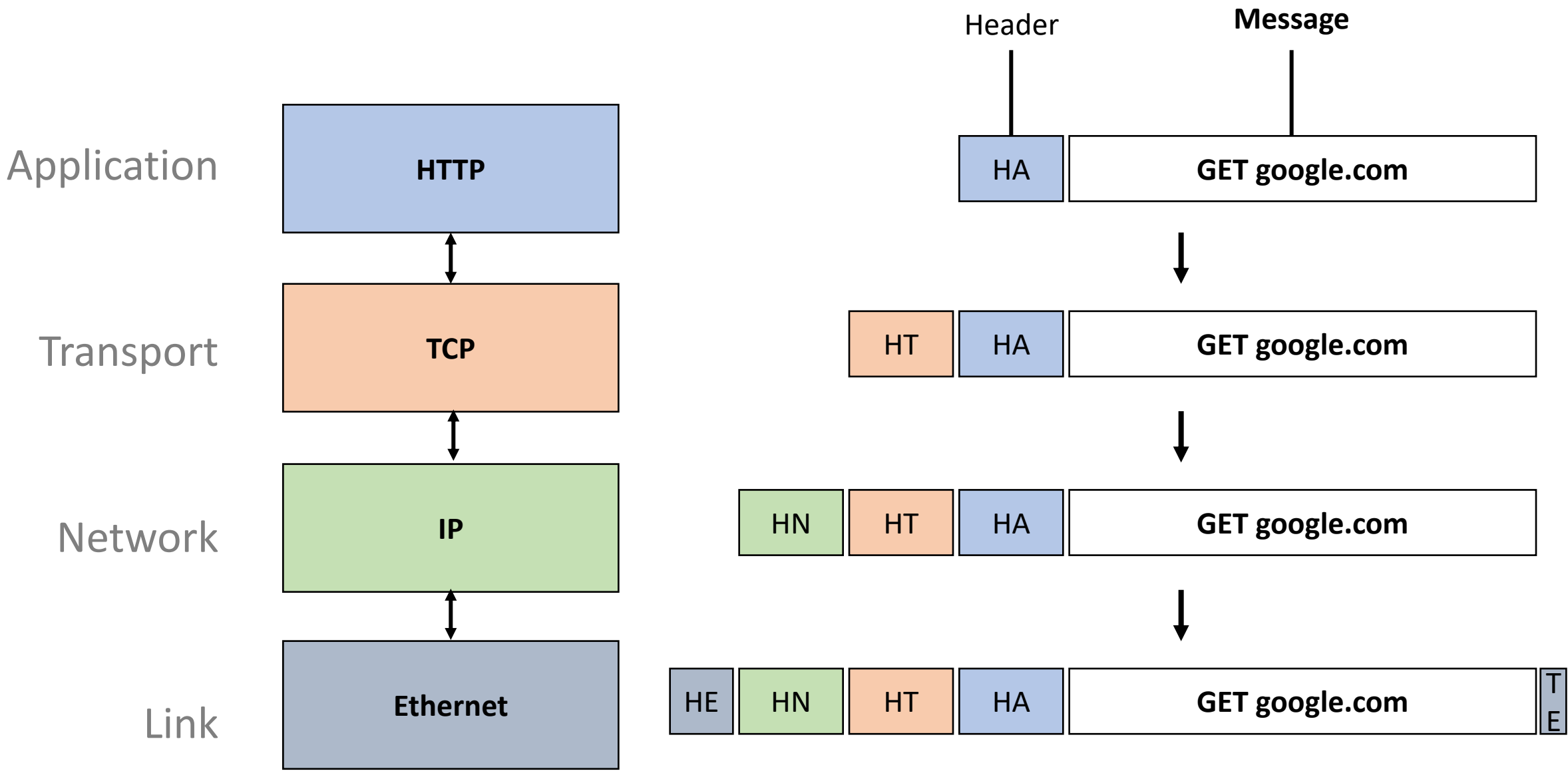

Application
HTTP
Transport
TCP
Network
IP
Link
Ethernet


Header
Message
HA
GET google.com

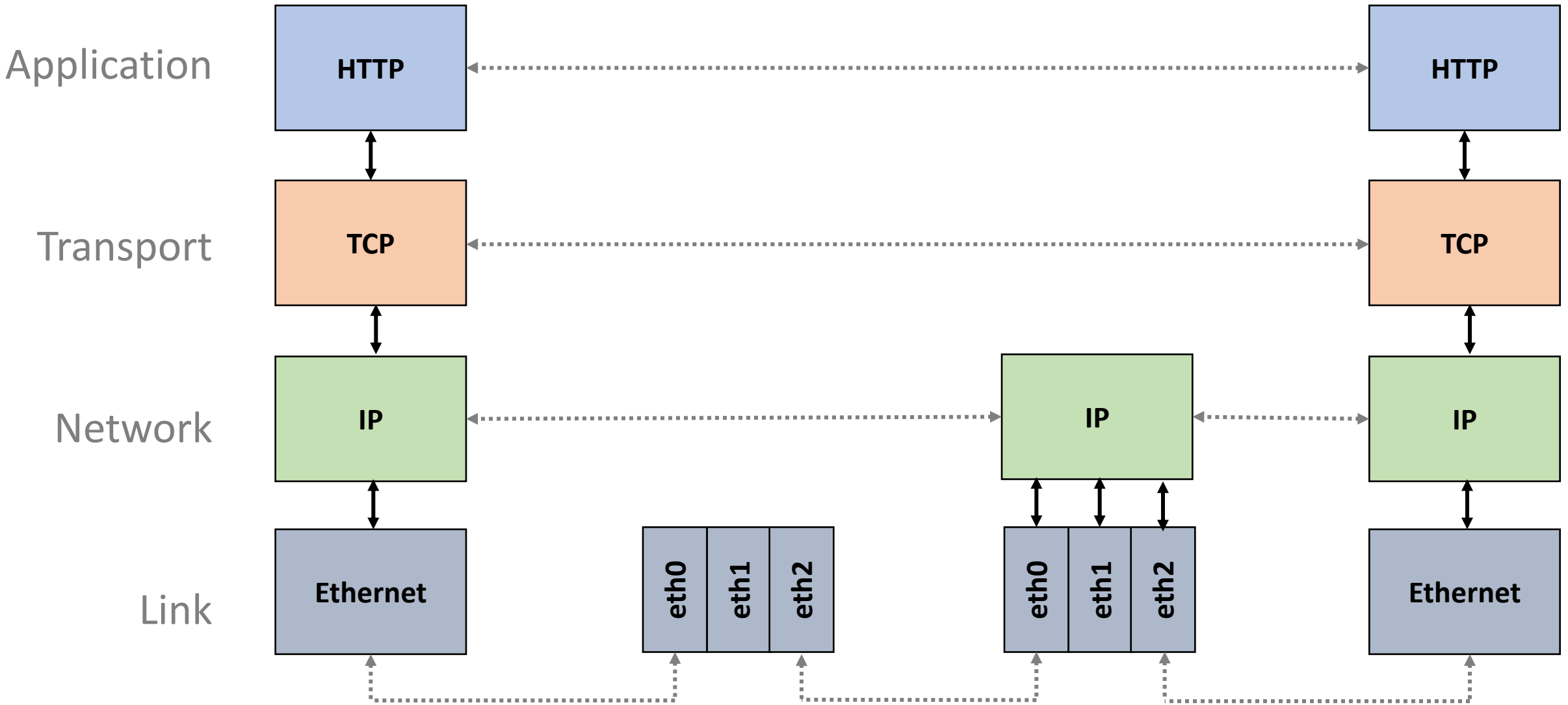

Application
HTTP
Transport
TCP
Network
IP
Link
Ethernet


Header
Message
HA
GET google.com
HT
HA
GET google.com

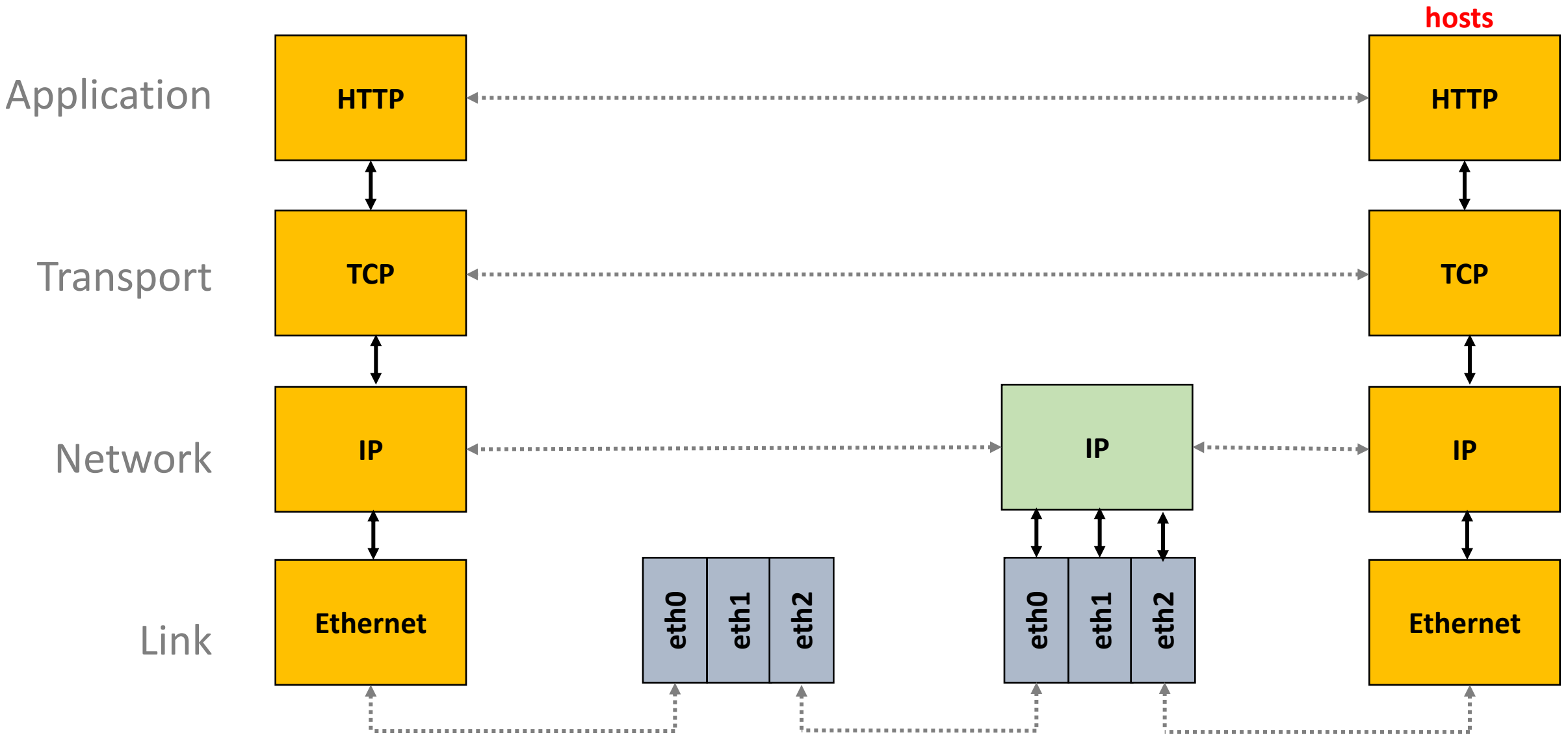

Application
Transport
Network
Link
HTTP
TCP
IP
Ethernet
Header
Message
HA
GET google.com
HT
HA
GET google.com
HN
HT
HA
GET google.com


Application
Transport
Network
Link
HTTP
TCP
IP
Ethernet
Header
Message
HA
GET google.com
HT
HN
HE
TE

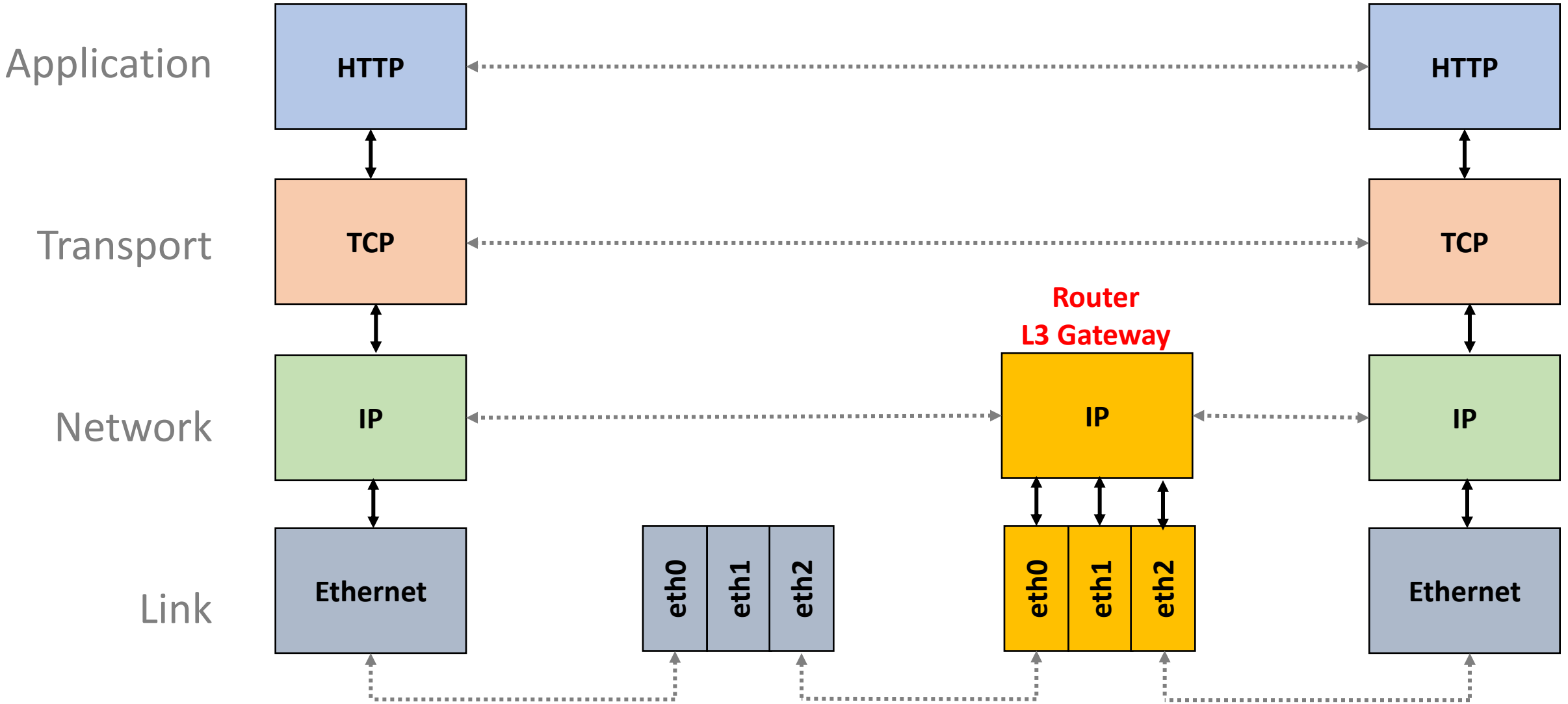
In practice, layers are **distributed** on every network device



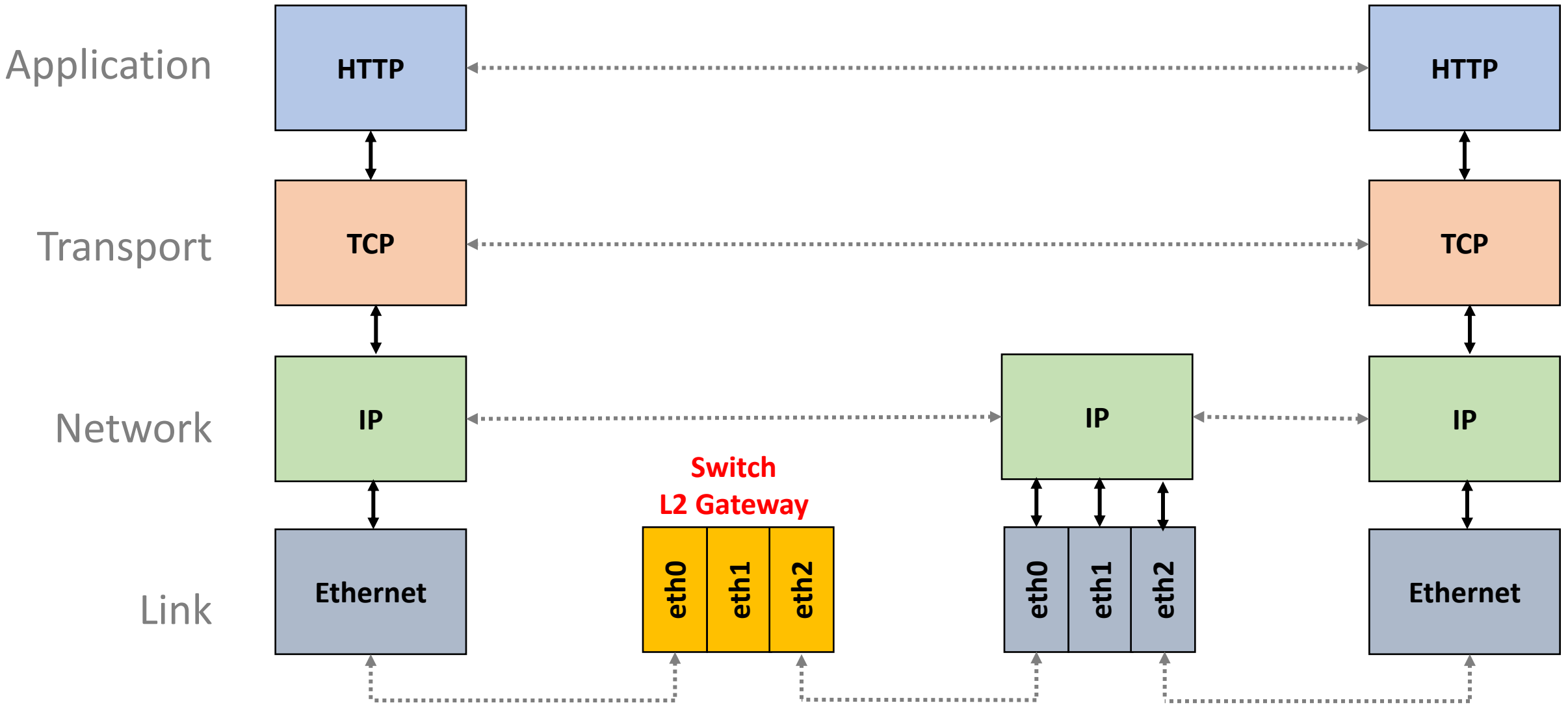
Since when bits arrive they must make it to the application, all the layers exist on a host



Routers act as L3 gateway
as such they implement L2 and L3



Switches act as L2 gateway
as such they only implement L2



Overview

How do we characterize the network?

A network *connection* is characterized by its delay, loss rate and throughput







How long does it take for a packet to reach the destination

What fraction of packets sent to a destination are dropped?

At what rate is the destination receiving data from the source?

Delay



Each packet suffers from several types of delays at *each node* along the path

transmission delay
\+ **propagation** delay
\+ **processing** delay
\+ **queueing** delay

due to **link properties**

due to **traffic mix & switch internals**

= **total** delay

Each packet suffers from several types of delays at *each node* along the path

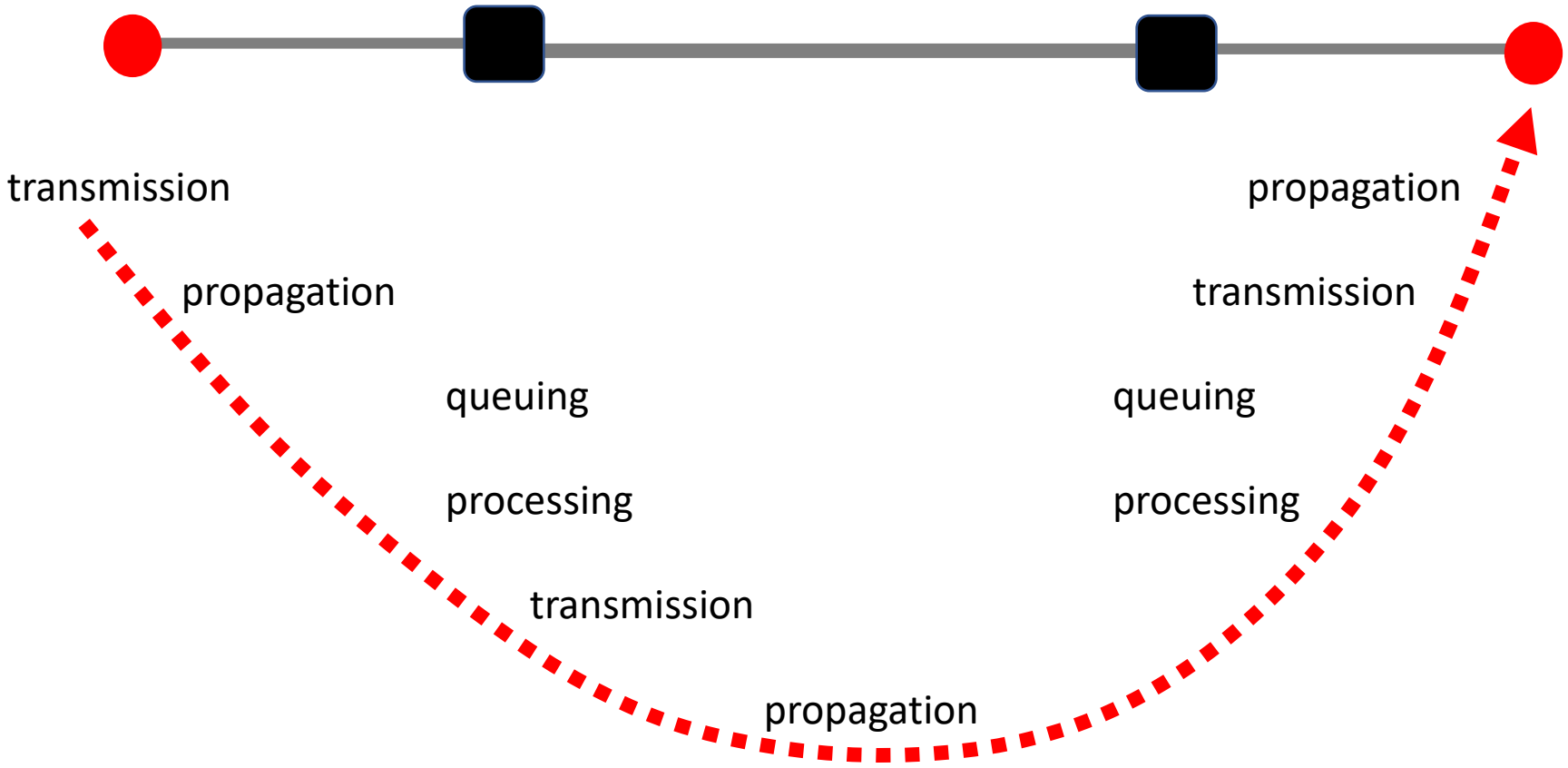
transmission delay

\+ **propagation** delay

~~\+ **processing** delay~~ **tend to be tiny**

\+ **queueing** delay

= **total** delay


transmission
propagation
queuing
processing
transmission
propagation
propagation
transmission
queuing
processing

The transmission delay is the amount of time required to push all of the bits onto the link

$$\text{Transmission delay [sec]} = \frac{\text{packet size [\#bits]}}{\text{link bandwidth [\#bits/sec]}}$$

$$\text{Example} = \frac{1000 \text{ bits}}{100 \text{ Mpbs}} = 10 \text{ µsec}$$

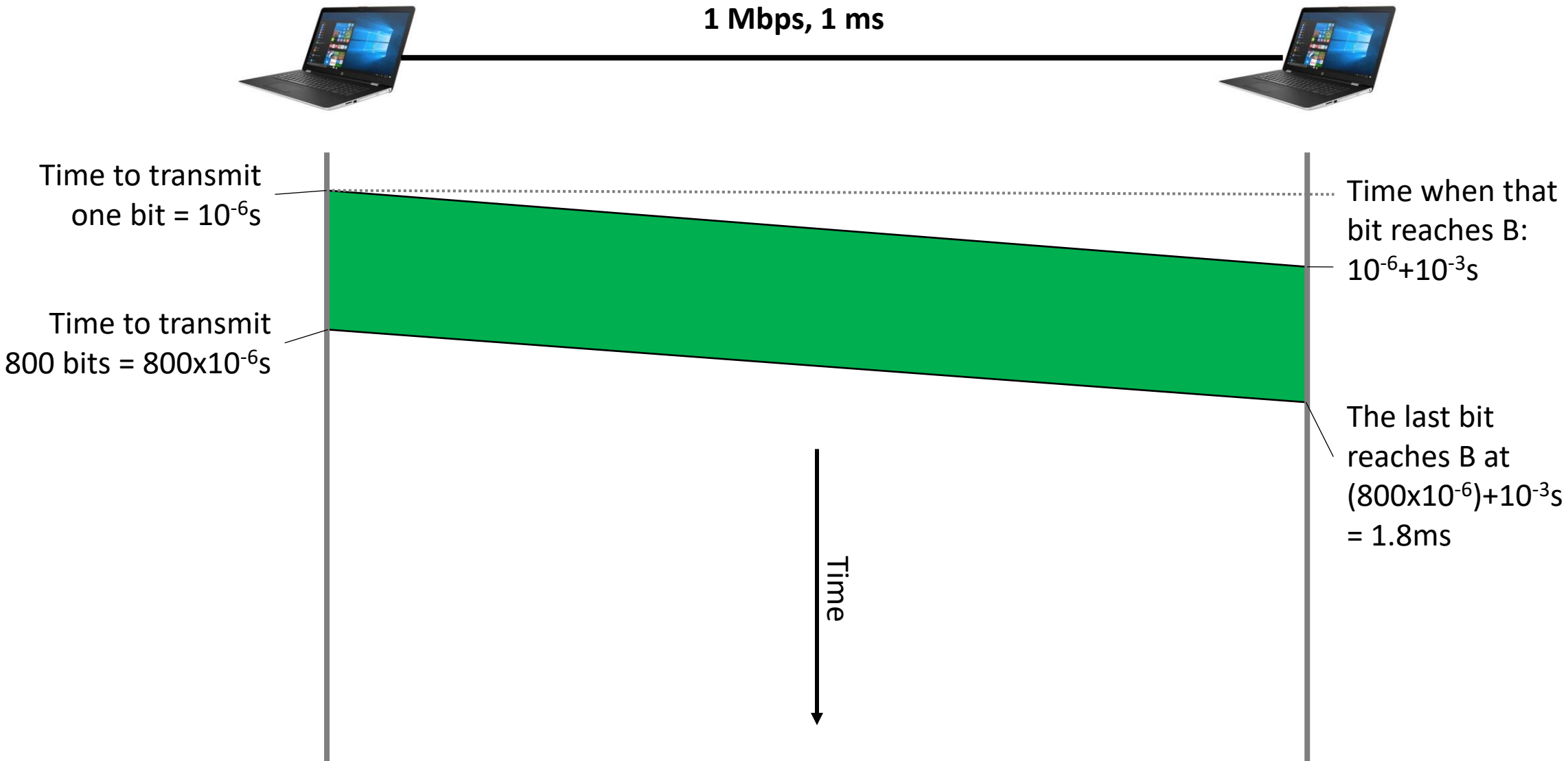
The propagation delay is the amount of time required for a bit to travel to the end of the link

$$\text{Propagation delay [sec]} = \frac{\text{link length [m]}}{\text{signal propagation speed [m/sec]}}$$

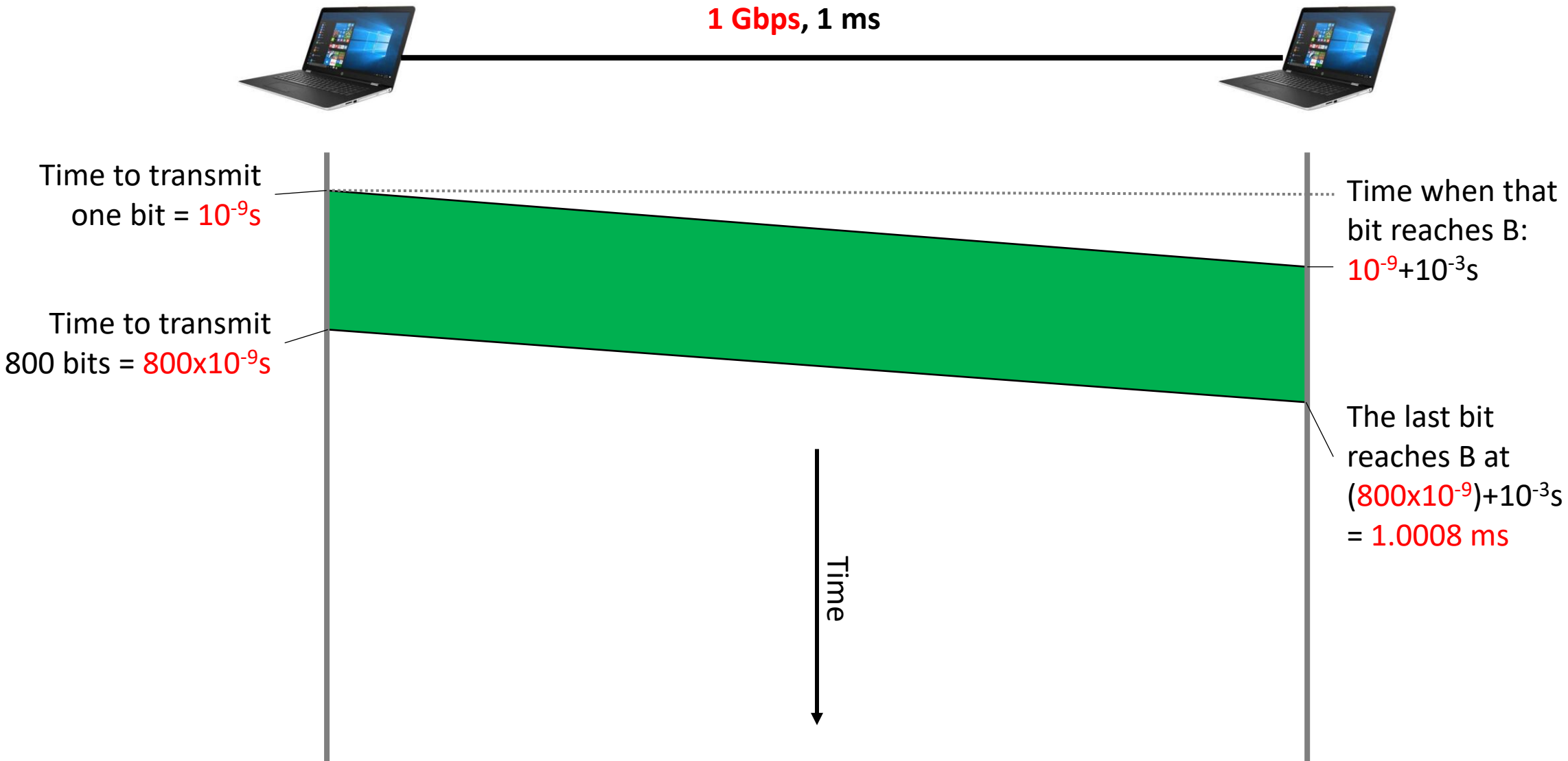
$$\text{Example} = \frac{30000 \text{ m}}{2 \times 10^8 \text{ m/sec}} = 150 \text{ µsec}$$

(speed of light in fiber)

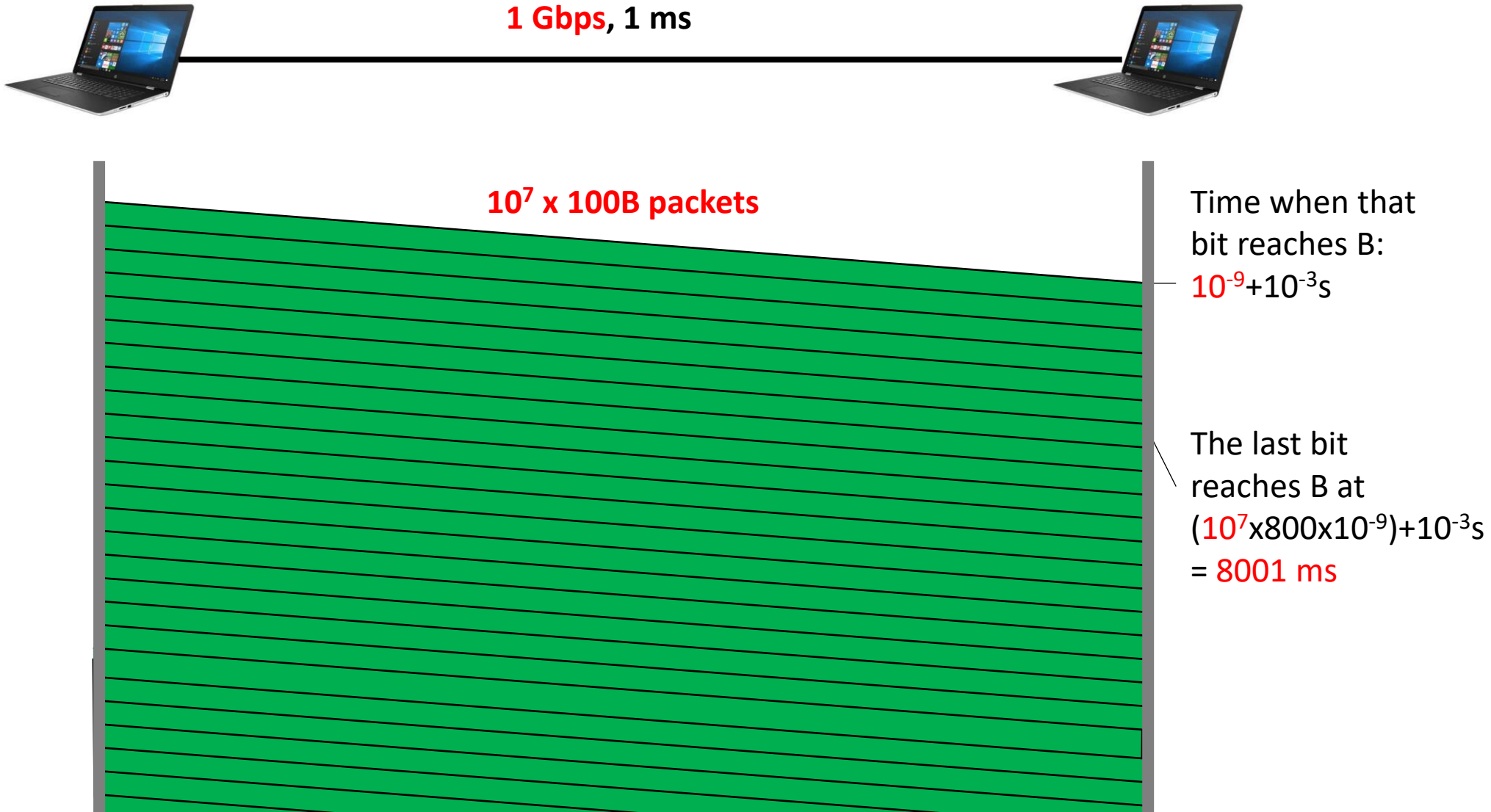
How long does it take to exchange 100 Bytes packet?



If we have a 1 Gbps link, the total time decreases to 1.0008 ms



If we now exchange a 1GB file split in 100B packets



Different transmission characteristics imply different tradeoffs in terms of which delay dominates

10^7x100B pkts	1Gbps link	***transmission delay dominates***
1x100B pkt	1Gbps link	***propagation delay dominates***
1x100B pkt	1Mbps link	***both matter***

In the Internet, we cannot know in advance which one matter!

The queuing delay is the amount of time a packet **waits** (in a buffer) to be transmitted on a link

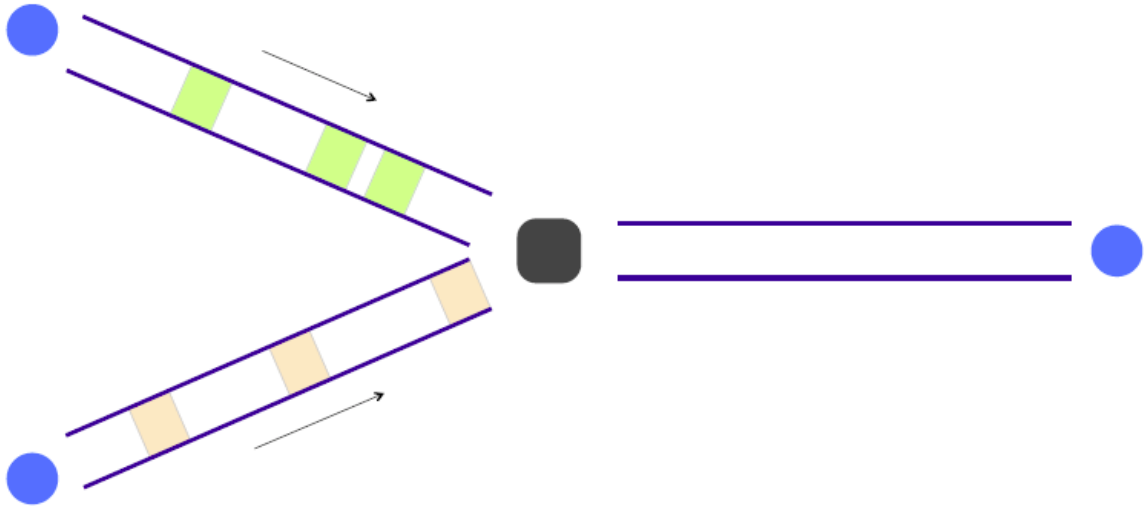
Queuing delay is the hardest to evaluate

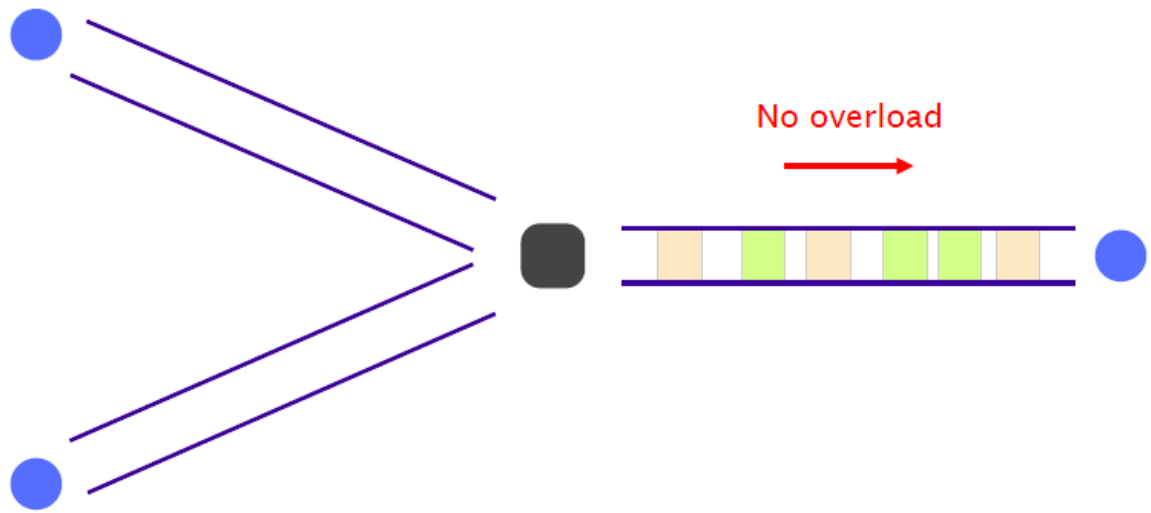
as it varies from packet to packet

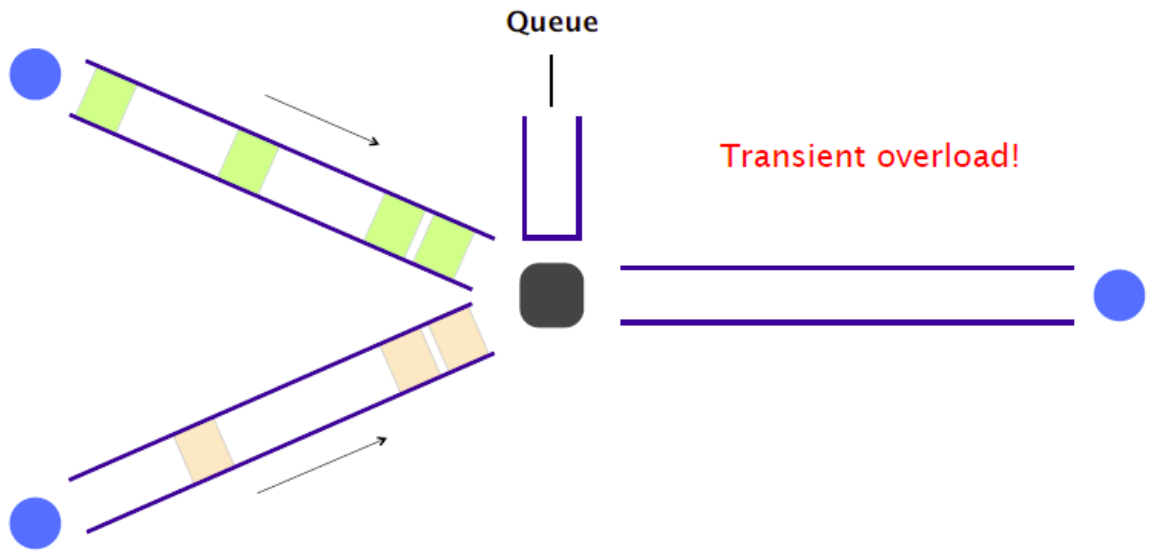
It is characterized with statistical measures

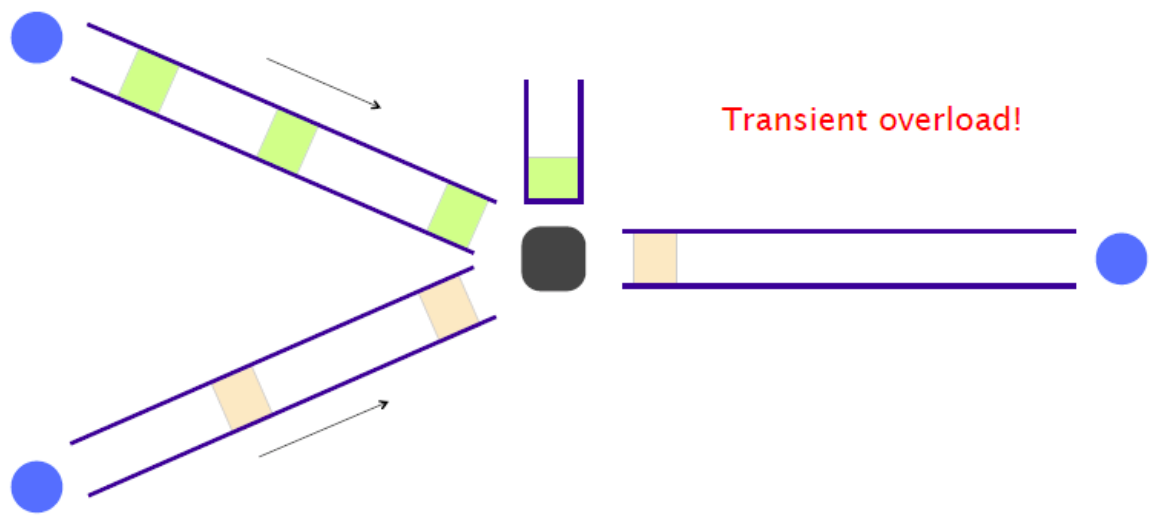
e.g., average delay & variance, probability of exceeding *x*

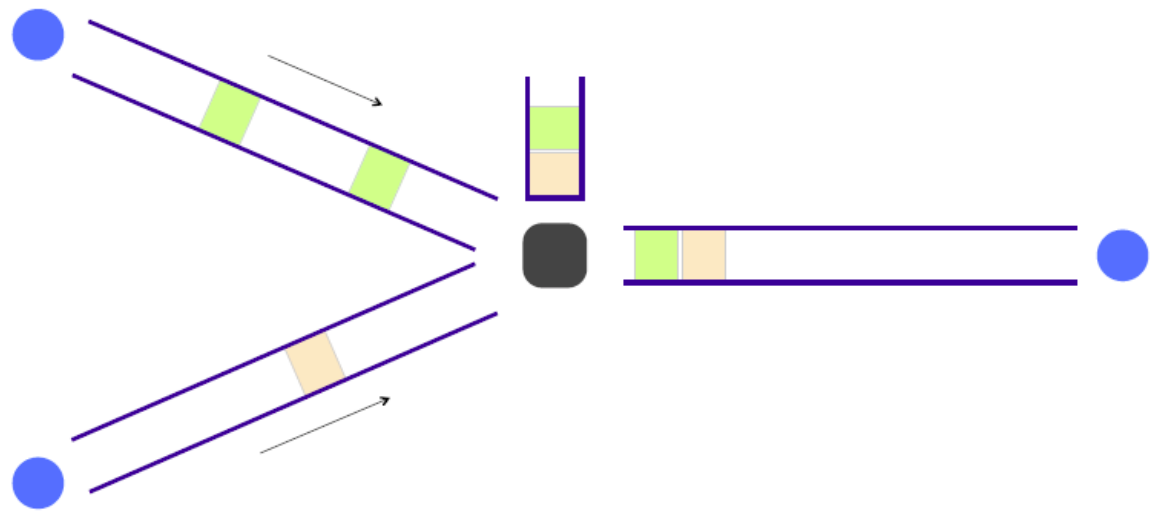
Queuing delay depends on the traffic pattern

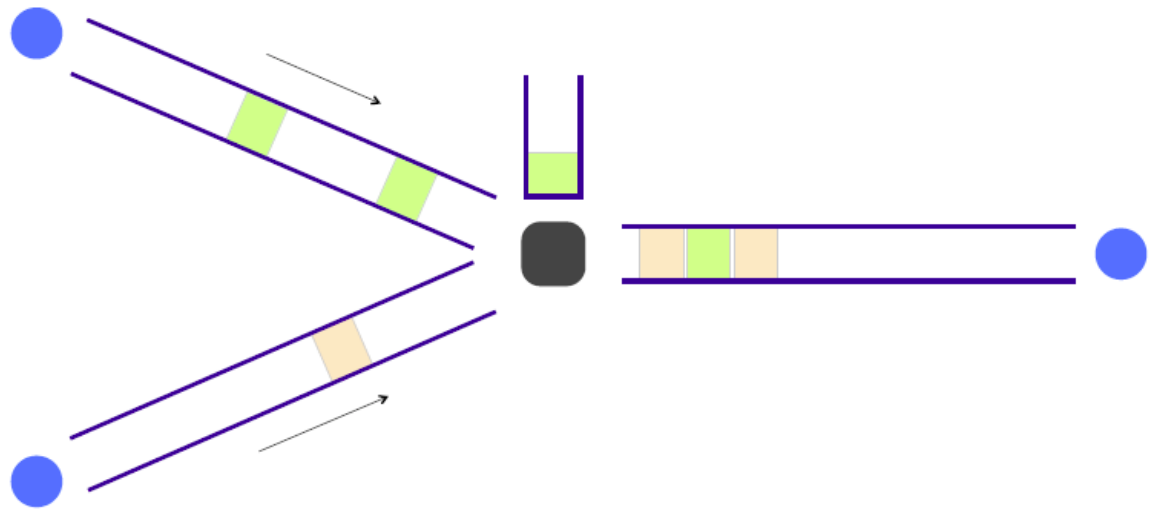


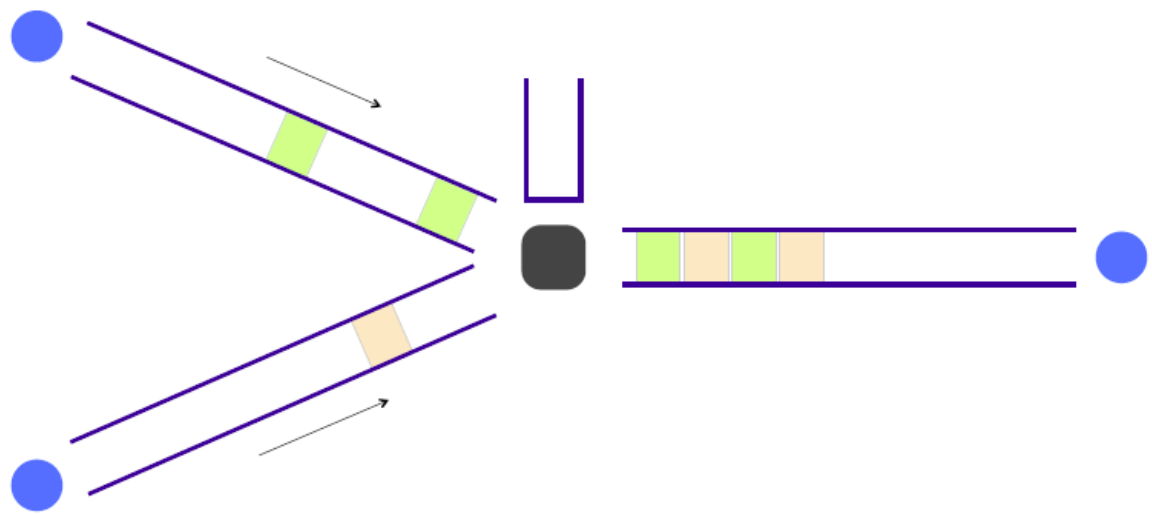

No overload


Queue
Transient overload!

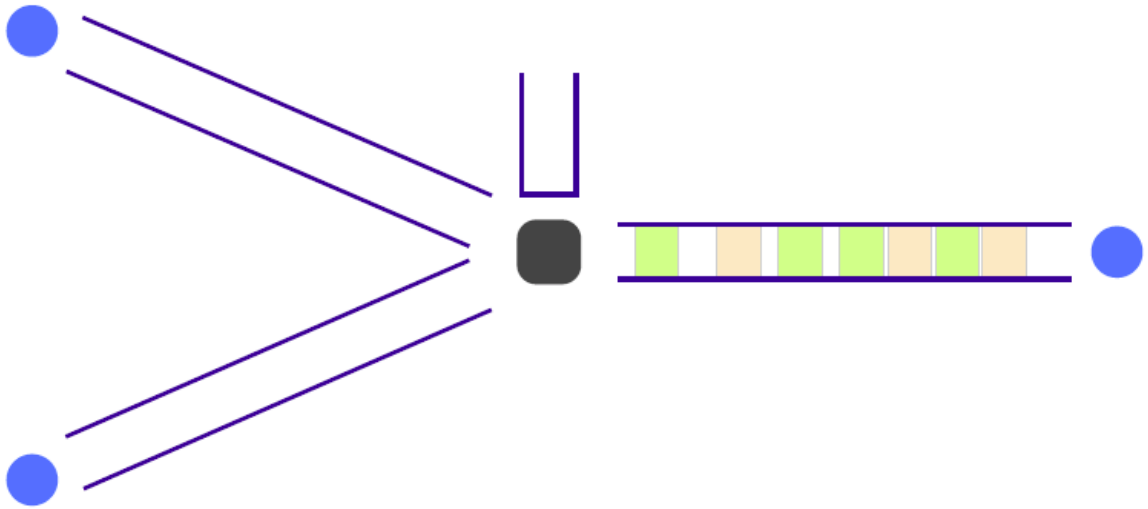

Transient overload!







Queues absorb transient bursts, but introduce queueing delays



The time a packet has to sit in a buffer before being processed depends on the traffic pattern

Queueing delay depends on:

arrival rate at the queue

transmission rate of the outgoing link

traffic **burstiness**

• **average packet arrival rate**	a	[packet/sec]
• **transmission rate of outgoing link**	R	[bit/sec]
• **fixed packets length**	L	[bit]
• **average bits arrival rate**	La	[bit/sec]
• **traffic intensity**	La/R	

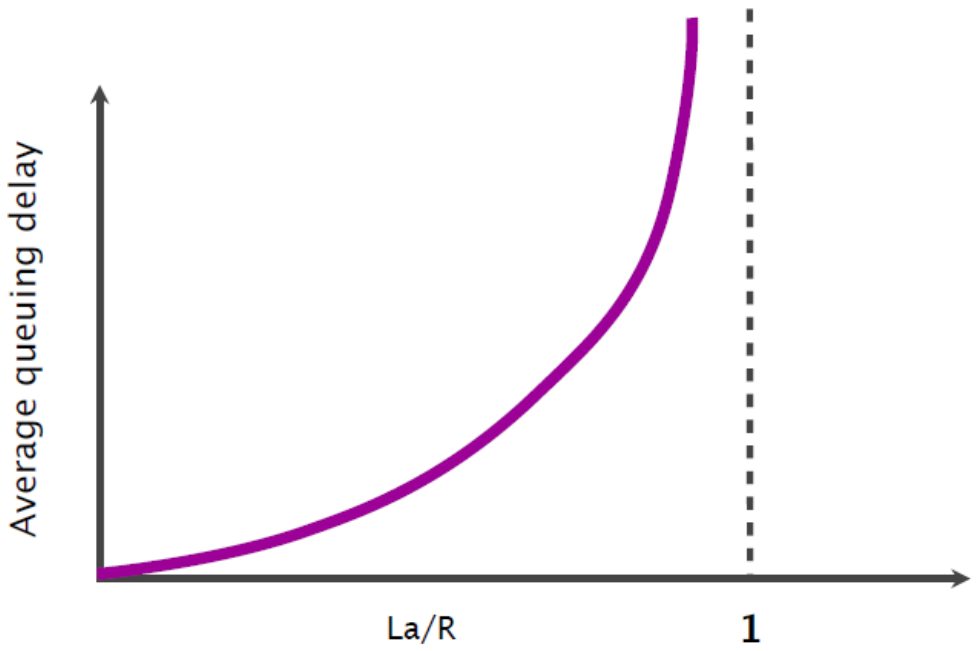
When the traffic intensity is >1, the queue will increase without bound, and so does the queuing delay

Golden rule

Design your queuing system,

so that it operates far from that point

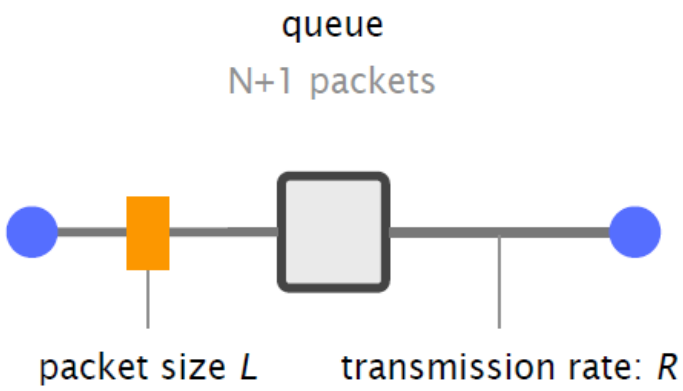
When the traffic intensity is <=1, queueing delay depends on the burst size



Loss

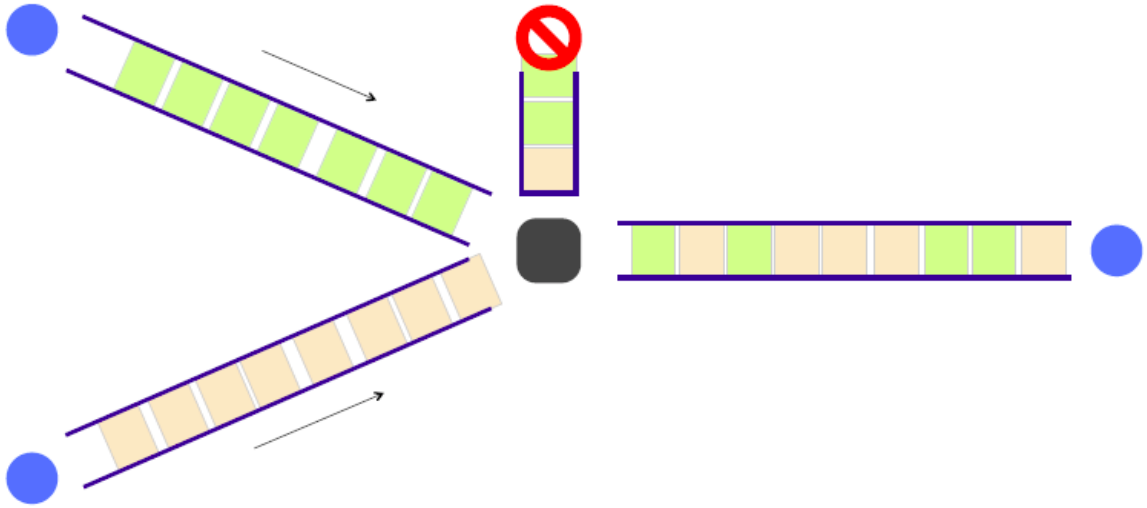


In practice, queues are not infinite.
There is an upper bound on queuing delay.



queuing delay upper bound: N*L/R

If the queue is persistently overloaded, it will eventually drop packets (loss)



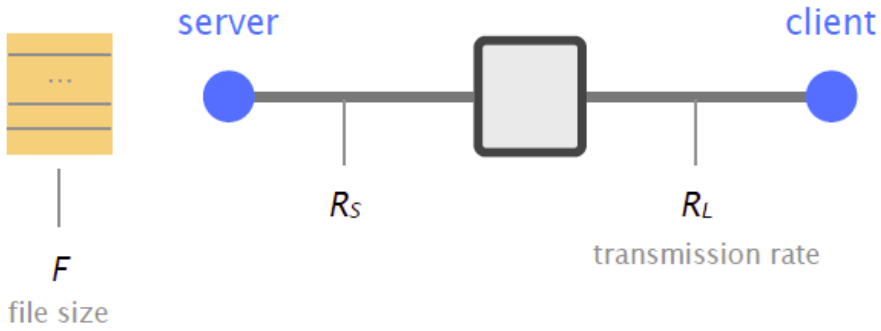
Throughput



The throughput is the instantaneous rate at which a host receives data

$$\text{Average throughput [\#bits/sec]} = \frac{\text{data size [\#bits]}}{\text{transfer time [sec]}}$$

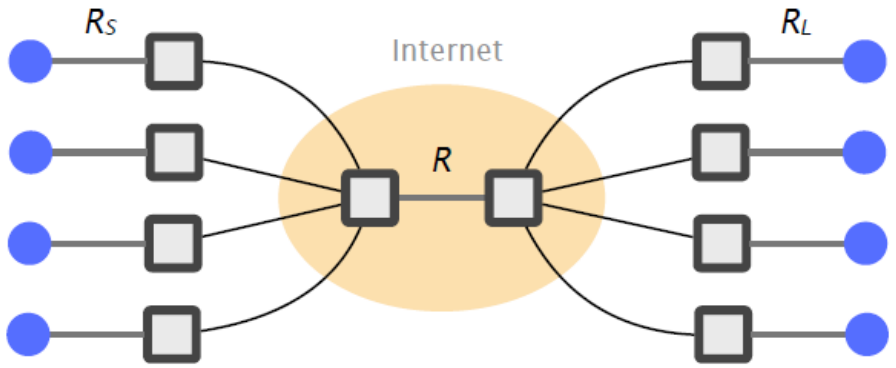
To compute throughput, one has to consider the bottleneck link



Average throughput $\min(R_S, R_L)$

= transmission rate of the bottleneck link

To compute throughput, one has to consider the bottleneck link… and the intervening traffic

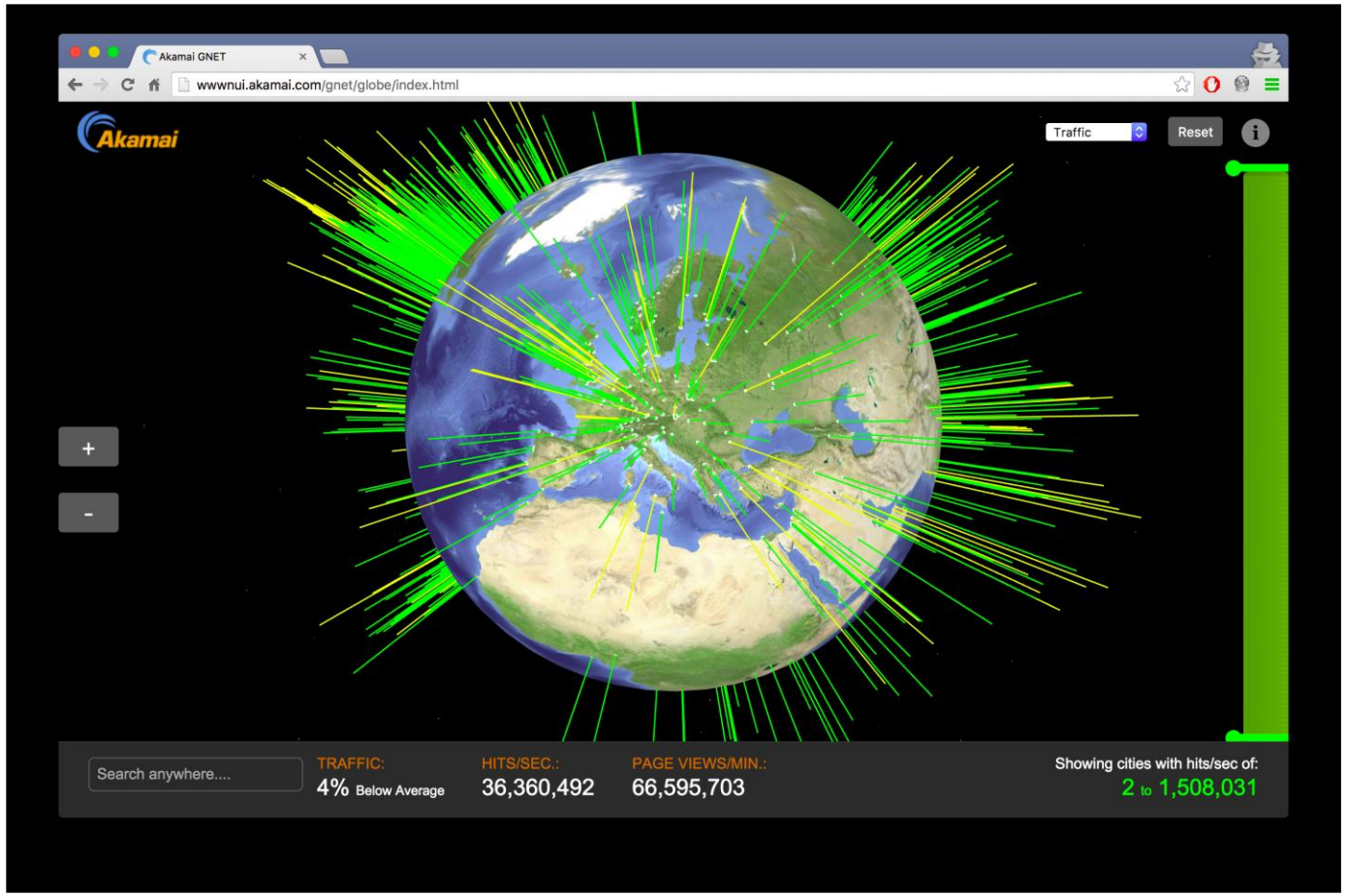


if 4*min(R_S,R_L) > R

the bottleneck is now in the core,
providing each download R/4 of throughput

As technology improves, throughput increase & delays are getting lower except for propagation (speed of light)

Because of propagation delays, Content Delivery Networks move content closer to you



* http://wwwnui.akamai.com/gnet/globe/index.html

To be continued…